

Ejercicios resueltos de ecuaciones diferenciales

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1 Ecuaciones diferenciales de primer orden

1.1 Ecuaciones lineales y reducibles a estas.

1.

$$\frac{dy}{dx} + 2y = 0$$

DEFINIMOS EL FACTOR INTEGRANTE.

$$p(x) = 2$$

factor integrante: $e^{\int 2dx} = e^{2x}$

multiplicamos la ecuacion por el factor integrante.

$$e^{2x} \frac{dy}{dx} + 2e^{2x}y = 0$$

el lado izquierdo de la ecuacion se reduce a:

$$\frac{d}{dx}[e^{2x}y] = 0$$

separamos variables e integramos.

$$\int \frac{d}{dx}[e^{2x}y] = 0 \int dx + c$$

$$e^{2x}y = c$$

$$y = ce^{-2x}$$

2.

$$\frac{dy}{dx} = 3y$$

forma lineal.

$$\frac{dy}{dx} - 3y = 0$$

$$p(x) = -3$$

Factor integrante: $e^{\int -3dx} = e^{-3x}$

multiplicamos por factor integrante.

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = 0$$

$$\int \frac{dy}{dx} [e^{-3x} y = 0] \int dx + c$$

$$e^{-3x} y = c$$

$$y = ce^{3x}$$

3.

$$3 \frac{dy}{dx} + 12y = 4$$

pasamos la ecuacion a la forma lineal.

$$\frac{dy}{dx} + 4y = \frac{4}{3}$$

$$p(x) = 4$$

Factor integrante: $e^{\int 4dx} = e^{4x}$

$$e^{4x} \frac{dy}{dx} + 4e^{4x} y = \frac{4}{3} e^{4x}$$

$$\int \frac{d}{dx} [e^{4x} y] = \int e^{4x} dx + c$$

$$e^{4x} y = \frac{1}{4} e^{4x} + c$$

$$y = \frac{1}{4} + ce^{-4x}$$

4.

$$y' = 2y + x^2 + 5$$

forma lineal

$$y' - 2y = x^2 + 5$$

Factor integrante: $e^{\int -2dx} = e^{-2x}$

$$e^{-2x} y' - 2e^{-2x} y = e^{-2x} x^2 + 5e^{-2x}$$

$$\int \frac{d}{dx} [e^{-2x} y] = \int e^{-2x} x^2 + 5 \int e^{-2x} + c$$

$$e^{-2x} y = -\frac{5}{2} e^{-2x} - \frac{1}{4} e^{-2x} (2x^2 + 2x + 1) + C$$

$$y = -\frac{x^2}{2} - \frac{x}{2} - \frac{1}{4} + \frac{5}{2} + ce^{2x}$$

5.

$$ydx - 4(x + y^6)dy = 0$$

$$ydx = 4(x + y^6)dy$$

$$\frac{dx}{dy} = \frac{4(x+y^6)}{y}; \frac{dx}{dy} = \frac{4x}{y} + \frac{4y^6}{y}$$

definimos la forma lineal.

$$\frac{dx}{dy} - \frac{4x}{y} = 4y^5$$

Factor integrante: $e^{-4 \int \frac{1}{y} dy}$; $e^{-4 \log(y)}$; $e^{\log(y)^{-4}}$; $y^{-4} = \frac{1}{y^4}$

$$\frac{1}{y^4} \frac{dx}{dy} - \frac{1}{y^4} \frac{4x}{y} = \frac{1}{y^4} 4y^5$$

$$\frac{d}{dy} \left[\frac{1}{y^4} x \right] = 4y$$

$$\int \frac{d}{dy} \left[\frac{1}{y^4} x \right] = 4 \int y dy$$

$$\frac{1}{y^4} x = 2y^2 + C$$

$$x = 2y^6 + cy^4$$

6.

$$xy' + y = e^x$$

$$y' + \frac{1}{x}y = \frac{e^x}{x}$$

Factor integrante:

$$e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

$$xy' + \frac{x}{x}y = \frac{xe^x}{x}$$

$$\frac{d}{dx}[xy] = e^x$$

Integramos:

$$\int \frac{d}{dx}[xy] = \int e^x dx + c$$

$$xy = e^x + c$$

$$y = e^x x^{-1} + cx^{-1}$$

7.

$$x \frac{dy}{dx} + y = \frac{2}{y^2}$$

$$\frac{dy}{dx} + \frac{y}{x} = \frac{2}{xy^2} \dots (1)$$

hacemos la sustitucion: $u = y^{1-n}$ donde $n = -2$

$$u = y^{1-(-2)} = y^3; u^{1/3} = y$$

Derivamos esta ultima.

$$\frac{1}{3}u^{-2/3} \frac{du}{dx} = \frac{dy}{dx}$$

Sustituimos en la ecuacion diferencial 1.

$$\frac{1}{3}u^{-2/3}\frac{du}{dx} + \frac{u^{1/3}}{x} = \frac{2(u^{1/3})^2}{x}$$

Acomodamos a la forma lineal, multiplicando toda la ecuacion por $\frac{1}{3}u^{2/3}$.

$$\frac{du}{dx} + 3\frac{u}{x} = \frac{6}{x}$$

Esta es una ecuacion lineal. Definimos el factor integrante.

$$e^{3\int \frac{1}{x}dx} = e^{3\log x} = e^{\log x^3} = x^3$$

Multiplicamos por factor integrante.

$$x^3\frac{du}{dx} + 3x^3\frac{u}{x} = x^3\frac{6}{x}$$

$$\frac{d}{dx}[x^3u] = 6x^2$$

integramos.

$$\int \frac{d}{dx}[x^3u] = 6\int x^2 + c$$

$$x^3u = 2x^3 + c$$

$$u = 2 + cx^{-3}$$

Sustituimos $u = y^3$

$$y^3 = 2 + cx^{-3}$$

$$8. y^{1/2}\frac{dy}{dx} + y^{3/2} = 1; \text{ condicion } y(0) = 4$$

$$\frac{dy}{dx} + \frac{y^{3/2}}{y^{1/2}} = \frac{1}{y^{1/2}} \leftrightarrow \frac{dy}{dx} + y = y^{-1/2}$$

$$u = y^{1-n}; n = -1/2; u = y^{1-(-1/2)} = y^{3/2}$$

$$u^{2/3} = y$$

$$\frac{2}{3}u^{-1/3}\frac{du}{dx} = \frac{dy}{dx}$$

Sustituimos.

$$\frac{2}{3}u^{-1/3}\frac{du}{dx} + u^{2/3} = (u^{2/3})^{-1/2}$$

Multiplicamos la ecuacion por $\frac{2}{3}u^{1/3}$

$$\frac{du}{dx} + \frac{3}{2}u = \frac{3}{2}$$

La ecuacion se redujo a una lineal.

Factor integrante: $e^{\frac{3}{2}\int dx} = e^{\frac{3}{2}x}$

$$e^{\frac{3}{2}x} \frac{du}{dx} + e^{\frac{3}{2}x} \frac{3}{2}u = e^{\frac{3}{2}x} \frac{3}{2}$$

$$\frac{d}{dx}[e^{\frac{3}{2}x}u] = \frac{3}{2}e^{\frac{3}{2}x}$$

$$\int \frac{d}{dx}[e^{\frac{3}{2}x}u] = \int \frac{3}{2}e^{\frac{3}{2}x}dx + c$$

$$e^{\frac{3}{2}x}u = e^{\frac{3}{2}x} + c$$

$$u = 1 + ce^{-\frac{3}{2}x}$$

Sustituimos $u = y^{3/2}$

$$y^{3/2} = 1 + ce^{-\frac{3}{2}x} \leftarrow \text{Solucion general.}$$

Ahora aplicamos las condiciones iniciales. $y(0) = 4$

$$4^{3/2} = 1 + ce^{-\frac{3}{2}0}$$

$$8 - 1 = c$$

$$c = 7$$

Sustituimos el valor de c en la ecuacion general.

$$y^{3/2} = 1 + 7e^{-\frac{3}{2}x} \leftarrow \text{Solucion particular.}$$

9.

$$y' + \frac{2}{x}y = -2xy^2$$

$$u = y^{1-n}; \text{ donde } n = 2$$

entonces:

$$u = y^{1-2}; u = y^{-1}; u^{-1} = y$$

$$-u^{-2} \frac{du}{dx} = \frac{dy}{dx}$$

sustituimos en la ecuacion.

$$-u^{-2} \frac{du}{dx} + \frac{2}{x}u^{-1} = -2x(u^{-1})^2$$

multiplicamos por $-u^2$

$$\frac{du}{dx} - \frac{2}{x}u = 2x$$

esta es una ecuacion lineal con $p(x) = -\frac{2}{x}$

obtenemos el factor integrante.

$$e^{-2 \int \frac{1}{x} dx} = e^{\log x^{-2}} = x^{-2}$$

$$x^{-2} \frac{du}{dx} - x^{-2} \frac{2}{x}u = x^{-2} 2x$$

$$\frac{d}{dx}[x^{-2}u] = 2x^{-1}$$

integramos.

$$\int \frac{d}{dx}[x^{-2}u] = \int 2x^{-1}dx + c$$

$$x^{-2}u = 2 \log x + c$$

$$u = 2x^2 \log x + cx^2$$

sustituimos $u = y^{-1}$
y la solución es entonces:

$$y = \frac{1}{2x^2 \log x + cx^2}$$

10,

$$y' + xy = xy^{-1/2}$$

sea. $n = -1/2$

$$u = y^{1-n}; u = y^{1-(-1/2)}; u = y^{3/2}; y = u^{2/3}$$

$$\frac{dy}{dx} = \frac{2}{3}u^{-1/3}$$

sustituimos en la ecuacion.

$$\frac{2}{3}u^{-1/3} + xu^{2/3} = x(u^{2/3})^{-1/2}$$

multiplicamos por $\frac{2}{3}u^{1/3}$

$$\frac{du}{dx} + \frac{3}{2}xu = \frac{3}{2}x \leftarrow \text{que es una ecuacion lineal con } p(x) = \frac{3}{2}x$$

Factor integrante:

$$e^{\frac{3}{2} \int x dx} = e^{\frac{3}{4}x^2}$$

$$e^{\frac{3}{4}x^2} \frac{du}{dx} + e^{\frac{3}{4}x^2} \frac{3}{2}xu = e^{\frac{3}{4}x^2} \frac{3}{2}x$$

$$\frac{d}{dx} e^{\frac{3}{4}x^2} u = \frac{3}{2} x e^{\frac{3}{4}x^2} dx + c$$

$$\int \frac{d}{dx} e^{\frac{3}{4}x^2} u = \frac{3}{2} \int x e^{\frac{3}{4}x^2} dx + c$$

$$e^{\frac{3}{4}x^2} u = e^{\frac{3}{4}x^2} + c$$

$$u = 1 + ce^{-\frac{3}{4}x^2}$$

sustituimos $u = y^{3/2}$

$$y^{3/2} = 1 + ce^{-\frac{3}{4}x^2}$$

1.2 Ecuaciones exactas y reducibles a exactas.

$$1.(2x-1)dx + (3y+1)dy = 0$$

$$M(x, y) = 2x - 1; N(x, y) = 3y + 1$$

Comprobamos que la ecuacion sea exacta, esto es si se cumple la condicion $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\frac{\partial M}{\partial y} = 0; \frac{\partial N}{\partial x} = 0$$

son iguales, por lo tanto la ecuacion es exacta.

Ahora tomamos una funcion $f_x(x, y) = M(x, y)$

$$f_x(x, y) = 2x - 1$$

integraremos respecto a x, y la constante de integracion sera una funcion $g(y)$

$$\int \frac{\partial M}{\partial x} = 2 \int x dx - \int dx + g(y)$$

$$f(x, y) = x^2 - x + g(y) \dots (1)$$

Esta funcion la derivamos con respecto de y.

$$\frac{\partial f}{\partial y} = g'(y)$$

igualamos con N(x,y)

$$g'(y) = 3y + 1$$

integraremos respecto a y

$$\int g'(y) = 3 \int y dy + \int dy + c$$

$$g(y) = \frac{3}{2}y^2 + y + c$$

sustituimos la funcion en (1).

$$x^2 - x + \frac{3}{2}y^2 + y = c$$

esta es una solucion en forma implicita de la ecuacion.

2.

$$(\sen y - y \sen x)dx + (\cos x + x \cos y - y)dy = 0$$

$$M(x, y) = \sen y - y \sen x; N(x, y) = \cos x + x \cos y - y$$

$$\frac{\partial M}{\partial y} = \cos y - \sen x$$

$$\frac{\partial N}{\partial x} = -\sen x + \cos y$$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ por lo tanto es una ecuacion exacta.

tomamos $f_x(x, y) = \text{sen}y - y\text{sen}x$

integraremos con respecto a x

$$\int f_x(x, y)dx = \int (\text{sen}y - y\text{sen}x)dx$$

$$f(x, y) = x\text{sen}y - y(-\cos x) + g(y) \dots (1)$$

derivamos esta ecuacion respecto a y, e igualamos con N(x,y)

$$f_y(x, y) = \cos x + x\cos y + g'(y) = \cos x + x\cos y - y$$

$$g'(y) = -y$$

integraremos respecto de y

$$\int g'(y) = - \int y dy + c$$

$$g(y) = -\frac{1}{2}y^2 + c$$

sustituimos en (1)

$$f(x, y) = x\text{sen}y + y\cos x - \frac{1}{2}y^2$$

nos queda la solucion implicita.

$$x\text{sen}y + y\cos x - \frac{1}{2}y^2 = c$$

3.

$$(3x^2y + e^y)dx = -(x^3 + xe^y - 2y)dy$$

$$M(x, y) = 3x^2y + e^y; N(x, y) = x^3 + xe^y - 2y$$

$$M_y(x, y) = 3x^2 + e^y$$

$$N_x(x, y) = 3x^2 + e^y$$

$M_y(x, y) = N_x(x, y)$ entonces es una ecuacion diferencial exacta.

Integramos $f_x(x, y)$ con respecto de x, y obtenemos una funcion $g(y)$ de constante de integracion.

$$f(x, y) = \int (3x^2y + e^y)dx$$

$$f(x, y) = x^3y + xe^y + g(y) \dots (1)$$

Derivamos con respecto de y (1) e igualamos con N(x,y)

$$f_y(x, y) = x^3 + xe^y + g'(y) = x^3 + xe^y - 2y$$

$$g'(y) = -2y$$

Integramos respecto de y

$$g(y) = -2 \int y dy + c$$

$$g(y) = -y^2 + c$$

sustituimos en (1)

$$x^3y + xe^y - y^2 = c... \text{ solucion implicita.}$$

4.

$$(6xy - 2y^2)dx + (3x^2 - 4xy)dy = 0$$

$$M_y(x, y) = 6x - 4y, N_x(x, y) = 6x - 4y$$

la ecuacion es exacta.

integramos $f_x(x, y)$ respecto a x.

$$f(x, y) = \int (6xy - 2y^2)dx$$

$$f(x, y) = 3x^2y - 2xy^2 + g(y)...(1)$$

derivamos respecto de y

$$f_y(x, y) = 3x^2 - 4xy + g'(y)$$

igualamos con $N(x, y)$

$$3x^2 - 4xy + g'(y) = 3x^2 - 4xy \rightarrow g'(y) = 0$$

integramos respecto de y

$$g(y) = c$$

sustituimos en la ecuacion (1)

$$3x^2y - 2xy^2 = c$$

5.

$$(2y - 2xy^3 + 4x + 6)dx + (2x - 3x^2y^2 - 1)dy = 0$$

con la condicion $y(-1) = 0$

$$M_y = 2 - 6xy^2 = N_x$$

Una vez comprobada que sea exacta.

integramos $f_x(x, y)$ respecto a x

$$f(x, y) = \int (2y - 2xy^3 + 4x + 6)dx$$

$$f(x, y) = 2xy - 3x^2y^3 + 2x^2 + 6x + g(y)...(1)$$

derivamos respecto a y :

$$f_x(x, y) = 2x - 3x^2y^2 + g'(y)$$

igualamos con $N(x, y)$

$$2x - 3x^2y^2 + g'(y) = 2x - 3x^2y^2 - 1 \rightarrow g'(y) = -1$$

integramos:

$$g(y) = -y + c$$

sustituimos en (1)

$$2xy - x^2y^3 + 2x^2 + 6x - y = c \dots \text{solucion implicita.}$$

$$\text{para } y(-1) = 0$$

$$2(-1)^2 + 6(-1) = c$$

$$c = -4$$

entonces la solucion particular al caso $y(-1)=0$ es:

$$2xy - x^2y^3 + 2x^2 + 6x - y = -4$$

6.

$$(-xy \sin x + 2y \cos x)dx + 2x \cos x dy = 0;$$

Use el factor integrante $\mu(x, y) = xy$

$$M_y(x, y) = -x \sin x + 2 \cos x$$

$$N_x(x, y) = -2x \sin x + 2 \cos x$$

$$N_x \neq M_y$$

la ecuacion es no exacta, en este ejemplo se nos dio el factor integrante, por lo tanto procedemos a multiplicar toda la ecuacion por el factor integrante.

$$xy(-xy \sin x + 2y \cos x)dx + xy(2x \cos x)dy = 0$$

$$(-x^2y^2 \sin x + 2xy^2 \cos x)dx + (2x^2y \cos x)dy = 0$$

comprobamos que esta ecuacion sea exacta.

$$M_y(x, y) = -2yx^2 \sin x + 4xy \cos x$$

$$N_X(x, y) = 4xy \cos x - 2x^2y \sin x$$

$M_Y = N_X$ por lo tanto esta ecuacion es exacta y la resolvemos como tal.

$$f_x(x, y) = -x^2 y^2 \sin x + 2xy^2 \cos x$$

integramos respecto a x:

$$f(x, y) = \int (-x^2 y^2 \sin x + 2xy^2 \cos x) dx$$

$$f(x, y) = x^2 y^2 \cos x + g(y) \dots (1)$$

derivamos respecto a y:

$$f_y(x, y) = 2x^2 y \cos x + g'(y)$$

igualamos con N_x

$$2x^2 y \cos x + g'(y) = 2x^2 y \cos x$$

$$g'(y) = 0$$

integramos respecto a y:

$$g(y) = c$$

sustituimos en (1)

$$f(x, y) = x^2 y^2 \cos x + c$$

2 Ecuaciones de orden superior

2.1 Ecuaciones diferenciales de orden superior reducibles a primer orden.

1. $y'' = 2x^2$

Integramos ambos lados de la ecuacion:

$$\int y'' = 2 \int x^2 dx + c$$

$$y' = \frac{2}{3}x^3 + c_1$$

Volvemos a integrar:

$$\int y' = \frac{2}{3} \int (x^3 + c_1) dx + c_2$$

$$y = (\frac{2}{3})(\frac{1}{4})x^4 + xc_1 + c_2$$

Solucion:

$$y = \frac{1}{6}x^4 + c_1x + c_2$$

$$2. \ y''' = \text{sen}(kx)$$

Integramos ambos lados de la ecuacion:

$$\int y''' = \int \text{sen}(kx)dx + c_1$$

$$y'' = -k\cos(kx) + c_1$$

$$\int y'' = -k \int \cos(kx)dx + c_1 \int dx + c_2$$

$$y' = -k^2 \text{sen}(kx) + xc_1 + c_2$$

$$\int y' = -k^2 \int \text{sen}(kx)dx + c_1 \int xdx + c_2 \int dx + c_3$$

$$y = k^3 \cos(kx) + \frac{1}{2}c_1x^2 + c_2x + c_3$$

$$3. \ y''' = \frac{1}{x}$$

Integrando:

$$\int y''' = \int \frac{1}{x}dx + c_1$$

$$y'' = \log x + c_1$$

$$\int y'' = \int \log x dx + c_1 \int dx + c_2$$

$$y' = x \log x - x + c_1x + c_2$$

$$\int y' = \int x \log x dx - \int x dx + c_1 \int x dx + c_2 \int dx + c_3$$

$$y = \frac{x^2}{2}(\log x - \frac{1}{2}) - \frac{1}{2}x^2 + c_1 \frac{1}{2}x^2 + c_2x + c_3$$

$$4. \ y'' = x + \sin x$$

Integrando:

$$\int y'' = \int x dx + \int \sin x dx + c_1$$

$$y' = \frac{1}{2}x^2 - \cos x + c_1$$

$$\int y' = \frac{1}{2} \int x^2 dx - \int \cos x dx + c_1 \int dx + c_2$$

$$y = \frac{1}{6}x^3 - \sin x + c_1x + c_2$$

$$5. \ y''' = x \sin x, \ y(0) = 0 \ y'(0) = 0 \ y''(0) = 2$$

Resolvemos la ecuacion diferencial integrando tres veces:

$$\int y''' = \int x \sin x dx + c_1$$

$$y'' = \sin x - x \cos x + c_1$$

$$\int y'' = \int \sin x dx - \int x \cos x dx + c_1 \int dx + c_2$$

$$y' = -\cos x - (\cos x + x \sin x) + c_1x + c_2$$

$$\int y' = -\int \cos x dx - \int \cos x dx - \int x \sin x dx + c_1 \int x dx + c_2 \int dx + c_3$$

$$y = -\sin x - \sin x - (-x \cos x + \sin x) + \frac{1}{2}c_1x^2 + c_2x + c_3$$

$$y = -3 \sin x + x \cos x + \frac{1}{2}c_1x^2 + c_2x + c_3$$

2.2 Reducibles a primer orden

1. $xy'' + y' = 0$

Definiendo:

$$p(x) = \frac{dy}{dx} \rightarrow \frac{dp}{dx} = \frac{d^2y}{dx^2}$$

$$xp' + p = 0$$

nos queda una ecuacion lineal homogenea de orden 1 de variables separables.

$$\frac{1}{x}dx = -\frac{1}{p}dp$$

$$\int \frac{1}{x}dx = -\int \frac{1}{p}dp + c_1$$

$$\log x = -\log p + \log c_1$$

$$\log x = \log\left(\frac{c_1}{p}\right)$$

Aplicando exponencial a ambos lados de la ecuacion.

$$x = \frac{c_1}{p}$$

hacemos $p(x) = \frac{dy}{dx}$

$$x = \frac{c_1}{dy/dx}$$

$$x = c_1 \frac{dx}{dy}$$

integrando:

$$\int \frac{1}{x}dx = \frac{1}{c_1} \int dy + c_2$$

$$\log x = \frac{1}{c_1}y + c_2$$

$y = c_1 \log x + c_2$. La constante de integracion conviene que tome valor positivo.

2. $(x-1)y'' - y' = 0$

Definimos:

$$p(x) = \frac{dy}{dx} \rightarrow \frac{dp}{dx} = \frac{d^2y}{dx^2}$$

$$(x-1)p' - p = 0$$

Dividimos entre $(x-1)$

$$\frac{x-1}{x-1}p' - \frac{1}{x-1}p = 0$$

$$p' - \frac{1}{x-1}p = 0$$

nos queda una ecuacion lineal homogenea.

$$\frac{dp}{dx} - \frac{1}{x-1}p = 0$$

$$\frac{dp}{dx} = \frac{1}{x-1}p$$

$$\frac{1}{p}dp = \frac{1}{x-1}dx$$

integrando:

$$\int \frac{1}{p}dp = \int \frac{1}{x-1}dx + c_1$$

$$\log(p) = \log(x-1) + \log(c_1)$$

$$\log(p) = \log[c_1(x-1)]$$

$$p = c_1(x-1)$$

$$\text{haciendo } p = \frac{dy}{dx}$$

$$\frac{dy}{dx} = c_1(x-1)$$

$$dy = c_1(x-1)dx$$

integrando:

$$\int dy = c_1 \int (x-1)dx + c_2$$

$$y = c_1 \frac{1}{2}x^2 - x + c_2$$

3.

2.3 Ecuaciones lineales homogeneas.

$$1. y'' + y' - 2y = 0$$

Resolvemos la ecuacion caracteristica asociada.

$$m^2 + m - 2 = 0$$

$$(m+2)(m-1) = 0$$

$$m_1 = -2 \quad m_2 = 1$$

Suponemos una solucion $y = e^{mx}$

$$y_1 = e^{-2x}$$

$$y_2 = e^x$$

$$y(x) = c_1 e^{-2x} + c_2 e^x$$

2.

$$y'' - 2y' + y = 0$$

Ecuacion caracteristica asociada $m^2 - 2m + 1 = 0$

$$(m-1)^2 = 0$$

$$m_{1,2} = 1$$

solucion $y = e^{mx}$

$$y_1 = e^x$$

$$y_2 = y_1 \int \frac{e^{\int p(y)dy}}{y_1^2} dx$$

$$y_2 = e^x \int \frac{e^{2x}}{e^{2x}} dx$$

$$y_2 = e^x x$$

solucion.

$$y(x) = c_1 e^x + c_2 x e^x$$

$$3. 4y'' - 8y' + 5y = 0$$

Ecuacion caracteristica.

$$4m^2 - 8m + 5 = 0$$

$$m_{1,2} = \frac{8 \pm \sqrt{64-80}}{8}$$

$$m_{1,2} = 1 \pm \frac{1}{2}i$$

solucion.

$$y = c_1 e^x e^{i\frac{1}{2}x} + c_2 e^x e^{-i\frac{1}{2}x}$$

$$y = e^x (c_1 e^{i\frac{1}{2}x} + c_2 e^{-i\frac{1}{2}x})$$

$$y = e^x (c_1 \cos \frac{1}{2}x + c_2 \sin \frac{1}{2}x)$$

$$4.3y'' - 2y' - 8y = 0$$

Ecuacion caracteristica:

$$3m^2 - 2y - 8 = 0$$

$$(3m + 4)(m - 2)$$

$$m_1 = 2$$

$$m_2 = -\frac{4}{3}$$

Solucion propuesta de la forma, $y = e^{mx}$

$$y_1 = e^{2x}$$

$$y_2 = -e^{-\frac{4}{3}x}$$

Solucion:

$$y(x) = c_1 e^{2x} + c_2 e^{\frac{4}{3}x}$$

$$5. y^v - 10y''' + 9y' = 0$$

Ecuacion caracteristica.

$$m^5 - 10m^3 + 9m = 0$$

$$m(m^4 - 10m^2 + 9) = 0$$

$$m_1 = 0 \quad (m^2 - 9)(m^2 - 1) \quad m_{2,3} = \pm 3 \quad m_{4,5} = \pm 1$$

Entonces tenemos las soluciones:

$$y_1 = e^0 = 1$$

$$y_2 = e^{3x}$$

$$y_3 = e^{-3x}$$

$$y_4 = e^x$$

$$y_5 = e^{-x}$$

Solucion:

$$y(x) = c_1 + c_2 e^{3x} + c_3 e^{-3x} + c_4 e^x + c_5 e^{-x}$$

$$6. y'' + 4y' + 3y = 0 \quad y(0) = 2 \quad y'(0) = -3$$

Ecuacion caracteristica.

$$m^2 + 4m + 3 = 0$$

$$m_{1,2} = \frac{-4 \pm \sqrt{-36}}{2}$$

$$m_{1,2} = -2 \pm 3i$$

Solucion:

$$y(x) = e^{-2x}(c_1 \cos 3x + c_2 \sin 3x)$$

$$y'(x) = e^{-2x}(-3c_1 \sin 3x + 3c_2 \cos 3x) - 2e^{-2x}(c_1 \cos 3x + c_2 \sin 3x)$$

Resolveremos para los casos $y(0) = 2$ y $y'(0) = -3$ particularmente.

Para $y(0) = 2$

$$2 = e^0(c_1 \cos 0 + c_2 \sin 0)$$

$$2 = c_1$$

Para $y'(0) = -3$

$$-3 = e^0(-3c_1 \sin 0 + 3c_2 \cos 0) - 2e^0(c_1 \cos 0 + c_2 \sin 0)$$

$$-3 = 3c_2 - 2c_1$$

$$-3 = 3c_2 - 2(2)$$

$$-3 + 4 = 3c_2$$

$$c_2 = \frac{1}{3}$$

Por lo tanto la solucion para el caso en general es:

$$y(x) = e^{-2x}(2 \cos 3x + \frac{1}{3} \sin 3x)$$

$$7 \cdot \frac{d^4 y}{dx^4} - 7 \frac{d^4 y}{dx^2} - 18y = 0$$

Ecuacion caracteristica:

$$m^4 - 7m^2 - 18 = 0$$

2.4 Ecuaciones no homogeneas de segundo orden

2.4.1 Coeficientes indeterminados.

1.

$$y'' + 3y' + 2y = 6$$

Resolvemos la ecuacion homogenea asociada

$$y_h = y'' + 3y' + 2y = 0$$

Ecuacion caracteristica:

$$m^2 + 3m + 2 = 0$$

$$(m - 1)(m - 2)$$

$$m_1 = 1 \quad m_2 = 2$$

$$y_h = c_1 e^x + c_2 e^{2x}$$

Ahora resolvemos la parte no homogenea suponiendo una solucion particular.

en este caso la parte no homogénea es 6, lo que nos sugiere usemos una solución de la forma A

$$y_p = A$$

$$y'_p = 0$$

$$y''_p = 0$$

Sustituimos en la ecuación original.

$$0 + 3(0) + 2A = 6$$

$$A = 3$$

Entonces la solución es $y(x) = y_h + y_p$

$$y(x) = c_1 e^x + c_2 e^{2x} + 3$$

$$2. \quad y'' + y = \sin x$$

Resolvemos primero la ecuación homogénea asociada.

$$y'' + y = 0$$

La ecuación característica de esta ecuación es.

$$m^2 + 1 = 0$$

$$m^2 = -1 \quad m_{1,2} = \pm \sqrt{-1} \quad m_{1,2} = \alpha \pm \beta i \text{ donde } \alpha = 0 \text{ y } \beta = 1$$

$$m_{1,2} = \pm i$$

$$y_h = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$$

$$y_h = c_1 \cos x + c_2 \sin x$$

Ahora buscamos una solución particular, para $\sin x$ nos proponen una solución de la forma $A \sin x + B \cos x$, sin embargo podemos observar que esta ya es una solución de la ecuación homogénea asociada $y'' + y = 0$, entonces según la regla de multiplicación para este caso, debemos multiplicar por x^n donde n es el número de enteros positivos que elimina la duplicación.

$$y_p = Ax \sin x + Bx \cos x$$

$$y'_p = A \sin x + Ax \cos x + B \cos x - Bx \sin x$$

$$y''_p = A \cos x + A \cos x - Ax \sin x - B \sin x - Bx \cos x - B \sin x = 2A \cos x - 2B \sin x - Ax \sin x - Bx \cos x$$

Sustituimos en la ecuación original

$$2A \cos x - 2B \sin x - Ax \sin x - Bx \cos x + Ax \sin x + Bx \cos x = \sin x$$

$$2A \cos x - 2B \sin x = \sin x$$

$$2A = 0 \text{ entonces } A = 0$$

$$-2B = 1 \text{ entonces } B = -\frac{1}{2}$$

Sustituyendo

$$y_p = -\frac{1}{2}x \cos x$$

$$y(x) = y_h + y_p$$

$$y(x) = c_1 \cos x + c_2 \sin x - \frac{1}{2}x \cos x$$

$$3. \quad y'' - 10y' + 25y = 30x + 3$$

Resolvemos la ecuación homogénea asociada.

$$m^2 - 10m + 25 = 0$$

$$m_{1,2} = 5$$

$$y_h = c_1 e^{5x} + c_2 x e^{5x}$$

La solución particular propuesta para $30x + 3$ es $Ax + B$

$$y_p = Ax + B$$

$$y'_p = A$$

$$y''_p = 0$$

sustituimos en la ecuación

$$-10(A) + 25(Ax + B) = 30x + 3$$

$$25A = 30 \dots (1) \text{ entonces } A = \frac{6}{5}$$

$$25B - 10A = 3 \dots (2)$$

$$25B - 10\left(\frac{6}{5}\right) = 3$$

$$25B = 3 + 12$$

$$B = \frac{3}{5}$$

$$y_p = \frac{6}{5}x + \frac{3}{5}$$

$$y(x) = y_h + y_p$$

$$y(x) = c_1 e^{5x} + c_2 x e^{5x} + \frac{6}{5}x + \frac{3}{5}$$

$$4. \quad \frac{1}{4}y'' + y' + y = x^2 - 2x$$

Resolvemos la ecuación homogénea asociada.

$$\frac{1}{4}y'' + y' + y = 0$$

$$\frac{1}{4}m^2 + m + 1 = 0$$

$$m_{1,2} = -2$$

$$y_h = c_1 e^{-2x} + c_2 x e^{-2x}$$

Ahora suponemos una solución particular para el caso de $f(x) = x^2 - 2x$

$$y_p = Ax^2 + Bx + C$$

$$y'_p = 2Ax + B$$

$$y''_p = 2A$$

Sustituimos en la ecuación original.

$$\frac{1}{4}(2A) + 2Ax + B + Ax^2 + Bx + C = x^2 - 2x$$

$$\frac{1}{2}A + B + Ax^2 + 2Ax + Bx + C = x^2 - 2x$$

$$A = 1$$

$$2A + B = 2$$

$$B = 2 - 2 = 0$$

$$\frac{1}{2}A + B + C = 0$$

$$\frac{1}{2}A + C = 0$$

$$C = -\frac{1}{2}A = -\frac{1}{2}$$

$$y_p = x^2 - \frac{1}{2}$$

$$y(x) = y_h + y_p$$

$$y(x) = c_1 e^{-2x} + c_2 x e^{-2x} + x^2 - \frac{1}{2}$$

$$5. \quad y'' + 3y = -48x^2 e^{3x}$$

Se resuelve la parte homogenea.

$$y'' + 3y = 0$$

$$m^2 + 3 = 0$$

$$m_{1,2} = \sqrt{-3} \quad m_{1,2} = \sqrt{3}i$$

$$y_h = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x$$

suponemos una solucion particular para $-48x^2 e^{3x}$

$$y_p = e^{3x}(Ax^2 + Bx + C)$$

$$y'_p = 3e^{3x}(Ax^2 + Bx + C) + e^{3x}(2Ax + B)$$

$$y''_p = 9e^{3x}(Ax^2 + Bx + C) + 3e^{3x}(2Ax + B) + 3e^{3x}(2Ax + B) + e^{3x}(2A) = 9e^{3x}(Ax^2 + Bx + C) + 3e^{3x}(4Ax + 2B) + e^{3x}(2A)$$

Susituimos en la ecuacion.

$$9e^{3x}(Ax^2 + Bx + C) + 3e^{3x}(4Ax + 2B) + e^{3x}(2A) + 9e^{3x}(Ax^2 + Bx + C) + 3e^{3x}(2Ax + B) = -48x^2 e^{3x}$$

$$9e^{3x} Ax^2 + 9e^{3x} Bx + 9e^{3x} C + 12e^{3x} Ax + 6e^{3x} B + 2e^{3x} A + 9e^{3x} Ax^2 + 9e^{3x} Bx + 9e^{3x} C + 6e^{3x} Ax + 3e^{3x} B = -48x^2 e^{3x}$$

$$9A + 9A = -48$$

$$18A = -48$$

$$A = -\frac{8}{3}$$

$$B = 0$$

$$C = 0$$

$$6.y'' - y' = -3$$

$$y'' - y' = 0$$

$$m^2 - m = 0$$

$$m(m - 1) = 0$$

$$m_1 = 0 \quad m_2 = 1$$

$$y_h = c_1 e^{0x} + c_2 e^x = c_1 + c_2 e^x$$

En este caso podemos ver claramente que existe ya una solución que es c_1 igual con -3

entonces por la regla de multiplicidad. la solución propuesta $y_p = Ax$

$$y_p = Ax$$

$$y'_p = A$$

$$y''_p = 0$$

Sustituyendo en la ecuación.

$$0 - A = -3 \text{ entonces, } A = 3$$

$$y_p = 3x$$

$$y(x) = y_h + y_p$$

$$y(x) = c_1 + c_2 e^x + 3x$$

$$7. \ y''' - 6y'' = 3 - \cos x$$

Ecuación homogénea asociada $y_h = y''' - 6y'' = 0$

$$m^3 - 6m^2 = 0$$

$$m^2(m - 6) = 0$$

$$m_{1,2} = 0 \quad m_3 = 6$$

$$y_h = c_1 + c_2 x + c_3 e^{6x}$$

La solución particular propuesta para $3 - \cos x$ es $y_{p1} = A$ $y_{p2} = B \cos x + C \sin x$ sin embargo en la solución y_{p1} se repite la constante, entonces la multiplicamos por x de acuerdo a la ley de multiplicidad nos queda $y_{p1} = Ax^2$

$$y_p = Ax^2 + B \cos x + C \sin x$$

$$y'_p = 2Ax - B \sin x + C \cos x$$

$$y''_p = 2A - B \cos x - C \sin x$$

$$y'''_p = B \sin x - C \cos x$$

Sustituyendo en la ecuación original.

$$B \sin x - C \cos x - 12A + 6B \cos x + 6C \sin x = 3 - \cos x$$

$$-12A = 3 ; A = -\frac{1}{4}$$

$$6B - C = 1 \dots (1)$$

$$6C + B = 0 \dots (2)$$

Iguando 1 y 2

$$B = \frac{6}{37}$$

$$C = \frac{1}{37}$$

$$y_p = \frac{1}{2}x^3 + \frac{6}{37}\cos x + \frac{1}{37}\sin x$$

$$y(x) = c_1 + c_2 x + c_3 e^{6x} - \frac{1}{4}x^2 + \frac{6}{37}\cos x + \frac{1}{37}\sin x$$

$$9.y'' + 2y' + y = \text{sen}x + 3\cos 2x$$

$$y_h = y'' + 2y' + y = 0$$

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0$$

$$m_{1,2} = -1$$

$$y_h = c_1 e^x + c_2 x e^x$$

Solucion particular

$$y_p = A\cos x + B\text{sen}x + C\cos 2x + D\text{sen}2x$$

$$y'_p = -A\text{sen}x + B\cos x - 2C\text{sen}2x + 2D\cos 2x$$

$$y''_p = -A\cos x - B\text{sen}x - 4C\cos 2x - 4D\text{sen}2x$$

sustituyendo.

$$-A\cos x - B\text{sen}x - 4C\cos 2x - 4D\text{sen}2x - 2A\text{sen}x + 2B\cos x - 4C\text{sen}2x + 4D\cos 2x + A\cos x + B\text{sen}x + C\cos 2x + D\text{sen}2x = \text{sen}x + 3\cos 2x$$

$$-3C\cos 2x - 3D\text{sen}2x - 2A\text{sen}x + 2B\cos x - 4C\text{sen}2x + 4D\cos 2x = \text{sen}x + 3\cos 2x$$

$$-3C + 4D = 3 \dots (1)$$

$$-3D - 4C = 0 \dots (2)$$

$$C = \frac{9}{25}$$

$$D = \frac{12}{25}$$

$$-2A = 1 ; A = -\frac{1}{2}$$

$$2B = 0 ; B = 0$$

$$y(x) = c_1 e^x + c_2 x e^x - \frac{1}{2}\cos x + \frac{9}{25}\cos 2x + \frac{12}{25}\text{sen}2x$$

2.5 Variacion de parametro.

1. $y'' + y = \sec x$

Resolvemos la parte homogenea de la ecuacion esta es $y_h = y'' + y = 0$

Para la ecuacion homogenea asociada, resolvemos la ecuacion caracteristica.

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m_{1,2} = \sqrt{-1} ; m_{1,2} = \pm i$$

$$m_{1,2} = \alpha \pm \beta i ; \text{donde } \alpha = 0 \beta = 1$$

$$y_h = c_1 \cos x + c_2 \sin x$$

Ahora identificamos $y_1 = \cos x$ y $y_2 = \sin x$ y las derivamos.

$$y_1 = \cos x \quad y_2 = \sin x$$

$$y_1' = -\sin x \quad y_2' = \cos x$$

A continuacion calculamos el Wronskiano:

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = [(\cos x)(\cos x)] - [(\sin x)(-\sin x)] = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = [(0)(\cos x)] - [(\sin x)(\sec x)] = -\sin x \sec x = -\frac{\sin x}{\cos x} = -\tan x$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = [(\cos x)(\sec x) - (0)(-\sin x)] = \cos x \sec x = \frac{\cos x}{\cos x} = 1$$

$$u_1' = \frac{W_1}{W} = \frac{-\tan x}{1} = -\tan x ; u_1 = -\int \tan x dx = -[-\ln(\cos x)] = \ln(\cos x)$$

$$u_2' = \frac{W_2}{W} = \frac{1}{1} = 1; u_2 = \int dx = x$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = \ln(\cos x) \cos x + x \sin x$$

$$y(x) = y_h + y_p$$

$$y(x) = c_1 \cos x + c_2 \sin x + \cos x \ln(\cos x) + x \sin x$$

$$2. \quad y'' + y = \sin x$$

Resolvemos $y_h = y'' + y = 0$

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m_{1,2} = \pm \sqrt{-1} ; m_{1,2} = \pm i$$

Donde:

$$\alpha = 0 \text{ y } \beta = 1$$

$$y_h = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$y_h = e^{0x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$y_h = c_1 \cos x + c_2 \sin x$$

Definimos y_1, y_2

$$y_1 = \cos x ; y_1' = -\sin x$$

$$y_2 = \sin x ; y_2' = \cos x$$

Calculamos el Wronskiano.

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sin x & \cos x \end{vmatrix} = -\sin^2 x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sin x \end{vmatrix} = \sin x \cos x$$

Ahora calculamos u_1, u_2 .

$$u_1' = -\frac{\sin^2 x}{1} = -\sin^2 x$$

$$u_1 = -\int \sin^2 x dx = \frac{x}{2} - \frac{1}{4} \sin 2x$$

$$u_2' = \frac{\sin x \cos x}{1} = \sin x \cos x$$

$$u_2 = \int \sin x \cos x dx = \frac{1}{2} \sin^2 x$$

$$y_p = u_1 y_1 + u_2 y_2 = \left(\frac{x}{2} - \frac{1}{4} \sin 2x\right) \cos x + \frac{1}{2} \sin^2 x (\sin x)$$

$$y_p = \frac{1}{2} x \cos x - \frac{1}{4} \cos x \sin 2x + \frac{1}{2} \sin^3 x$$

$$y(x) = y_p + y_h$$

$$y(x) = c_1 \cos x + c_2 \sin x + \frac{1}{2} x \cos x - \frac{1}{4} \cos x \sin 2x + \frac{1}{2} \sin^3 x$$

$$3. y'' + y = \cos^2 x$$

Ecuacion homogenea asociada $y_h = y'' + y = 0$

Esta ecuacion tiene solucion de la forma:

$$y_h = c_1 \cos x + c_2 \sin x$$

Definimos y_1, y_2

$$y_1 = \cos x ; y_1' = -\sin x$$

$$y_2 = \sin x ; y_2' = \cos x$$

Calculamos los Wronskianos:

$$W = \begin{vmatrix} \cos x & \operatorname{sen} x \\ -\operatorname{sen} x & \cos x \end{vmatrix} = \cos^2 x + \operatorname{sen}^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \operatorname{sen} x \\ \cos^2 x & \cos x \end{vmatrix} = -\operatorname{sen} x \cos^2 x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\operatorname{sen} x & \cos^2 x \end{vmatrix} = \cos^3 x$$

Definimos u_1, u_2

$$u_1' = \frac{-\operatorname{sen} x \cos^2 x}{1} = -\operatorname{sen} x \cos^2 x$$

$$u_1 = -\int \operatorname{sen} x \cos^2 x dx = -\left[-\frac{\cos^3 x}{3} \right] = \frac{\cos^3 x}{3}$$

$$u_2' = \frac{\cos^3 x}{1} = \cos^3 x$$

$$u_2 = \int \cos^3 x dx = \operatorname{sen} x - \frac{\operatorname{sen}^3 x}{3}$$

$$y_p = u_1 y_1 + u_2 y_2 = \left(\frac{\cos^3 x}{3} \right) (\cos x) + \left(\operatorname{sen} x - \frac{\operatorname{sen}^3 x}{3} \right) (\operatorname{sen} x)$$

$$y_p = \frac{\cos^4 x}{3} + \operatorname{sen}^2 x - \frac{\operatorname{sen}^4 x}{3}$$

$$y(x) = c_1 \cos x + c_2 \operatorname{sen} x + \frac{\cos^4 x}{3} + \operatorname{sen}^2 x - \frac{\operatorname{sen}^4 x}{3}$$

$$4. y'' - y = \cosh x$$

Ecuacion homogenea asociada $y'' - y = 0$

$$m^2 - 1 = 0$$

$$m^2 = 1; m_{1,2} = \pm \sqrt{1} = \pm 1$$

$$y_h = c_1 e^x + c_2 e^{-x}$$

Definimos y_1, y_2

$$y_1 = e^x; y_1' = e^x$$

$$y_2 = e^{-x}; y_2' = -e^{-x}$$

Calculamos los Wronskianos:

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^{-x}(e^x) - e^x(e^{-x}) = -1 - 1 = -2$$

$$W_1 = \begin{vmatrix} 0 & e^{-x} \\ \cosh x & -e^{-x} \end{vmatrix} = -e^{-x}(\cosh x) = -e^{-x} \cosh x$$

$$W_2 = \begin{vmatrix} e^x & 0 \\ e^x & \cosh x \end{vmatrix} = e^x \cosh x$$

Calculamos u_1 y u_2

$$u'_1 = \frac{-e^{-x} \cosh x}{-2} = \frac{1}{2} e^{-x} \cosh x$$

$$u_1 = \frac{1}{2} \int e^{-x} \cosh x dx = \frac{1}{8} e^{-2x} (2e^{2x} x - 1)$$

$$u'_2 = \frac{e^x \cosh x}{-2} = -\frac{1}{2} e^x \cosh x$$

$$u_2 = -\frac{1}{2} \int e^x \cosh x dx = -\frac{1}{2} \left[\frac{x}{2} + \frac{e^{2x}}{4} \right]$$

$$y_p = e^x \left[\frac{1}{8} e^{-2x} (2e^{2x} x - 1) \right] + (-e^{-x}) \left(-\frac{x}{4} - \frac{e^{2x}}{8} \right)$$

$$y_p = \frac{1}{8} e^{-x} (2e^{2x} x - 1) + \frac{x e^{-x}}{4} + \frac{e^x}{8}$$

$$y(x) = c_1 e^x + c_2 e^{-x} + \frac{1}{8} e^{-x} (2e^{2x} x - 1) + \frac{x e^{-x}}{4} + \frac{e^x}{8}$$

$$4. \quad y'' + 3y' + 2y = \frac{1}{1 + e^x}$$

Ecuacion homogenea asociada $y_h = y'' + 3y' + 2y = 0$

$$m^2 + 3m + 2 = 0$$

$$(m + 2)(m + 1) = 0$$

$$m_1 = -2 \quad m_2 = -1$$

$$y_h = c_1 e^{-2x} + c_2 e^{-x}$$

Definimos y_1, y_2 .

$$y_1 = e^{-2x}; \quad y'_1 = -2e^{-2x}$$

$$y_2 = e^{-x}; \quad y'_2 = -e^{-x}$$

Calculamos Wronskianos:

$$W = \begin{vmatrix} e^{-2x} & e^{-x} \\ -2e^{-2x} & -e^{-x} \end{vmatrix} = (e^{-2x})(-e^{-x}) - (e^{-x})(-2e^{-2x}) = -e^{-3x} + 2e^{-3x} = e^{-3x}$$

$$W_1 = \begin{vmatrix} 0 & e^{-x} \\ \frac{1}{1+e^x} & -e^{-x} \end{vmatrix} = -\frac{e^{-x}}{1+e^x}$$

$$W_2 = \begin{vmatrix} e^{-2x} & 0 \\ -2e^{-2x} & \frac{1}{1+e^x} \end{vmatrix} = \frac{e^{-2x}}{1+e^x}$$

Calculamos u_1, u_2

$$u'_1 = -\frac{\frac{e^{-x}}{1+e^x}}{\frac{1}{e^{-3x}}} = \frac{e^{-x}}{(e^{-3x})(1+e^x)} = \frac{1}{e^{-2x}(1+e^x)} = -\frac{1}{e^{-2x} + e^{-x}}$$

$$u_1 = -\int \frac{1}{e^{-2x} + e^{-x}} dx = -e^x + \ln(e^x + 1) - 1$$

$$u'_2 = \frac{\frac{e^{-2x}}{1+e^x}}{\frac{1}{e^{-3x}}} = \frac{e^{-2x}}{(e^{-3x})(1+e^x)} = \frac{1}{e^{-x} + 1}$$

$$u_2 = \int \frac{1}{e^{-x} + 1} dx = x + \ln(e^{-x} + 1)$$

$$y_p = (e^{-2x})[-e^x + \ln(e^x + 1) - 1] + [x + \ln(e^{-x} + 1)](e^{-x})$$

$$y_p = -e^{-x} + e^{-2x}\ln(e^x + 1) - e^{-2x} + xe^{-x} + e^{-x}\ln(e^{-x} + 1)$$

$$y(x) = c_1 e^{-2x} + c_2 e^{-x} - e^{-x} + e^{-2x}\ln(e^x + 1) - e^{-2x} + xe^{-x} + e^{-x}\ln(e^{-x} + 1)$$

$$5.3y'' - 6y' + 6y = e^x \sec x$$

$$y_h = 3y'' - 6y' + 6y = 0$$

$$3m^2 - 6m + 6 = 0$$

$$a = 3, b = -6, c = 6$$

Definimos y_1, y_2

$$m_{1,2} = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(6)}}{2(3)} = \frac{6}{6} \pm \frac{\sqrt{36 - 72}}{6} = 1 \pm \frac{\sqrt{-36}}{6} = 1 \pm i$$

$$\alpha = 1, \beta = 1$$

$$y_h = e^x(c_1 \cos x + c_2 \sin x)$$

Definiendo y_1, y_2

$$y_1 = e^x \cos x ; y_1' = e^x \cos x - e^x \sin x$$

$$y_2 = e^x \sin x ; y_2' = e^x \sin x + e^x \cos x$$

Calculamos los Wronskianos

$$W = \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x \cos x - e^x \sin x & e^x \sin x + e^x \cos x \end{vmatrix} = (e^x \cos x)(e^x \sin x + e^x \cos x) - (e^x \sin x)(e^x \cos x - e^x \sin x) = e^x (\cos x \sin x + \cos^2 x - \cos x \sin x + \sin^2 x)$$

$$W = e^x (\cos^2 x + \sin^2 x) = e^x$$

$$W_1 = \begin{vmatrix} 0 & e^x \sin x \\ e^x \sec x & e^x \sin x + e^x \cos x \end{vmatrix} = -(e^x \sin x)(e^x \sec x) = -e^x \left(\frac{\sin x}{\cos x} \right) = -e^x \tan x$$

$$W_2 = \begin{vmatrix} e^x \cos x & 0 \\ e^x \cos x - e^x \sin x & e^x \sec x \end{vmatrix} = (e^x \cos x)(e^x \sec x) = e^x \left(\frac{\cos x}{\cos x} \right) = e^x$$

Calculamos u_1, u_2

$$u_1' = -\frac{e^x \tan x}{e^x} = -\tan x$$

$$u_1 = -\int \tan x dx = -(-\ln \cos x) = \ln \cos x$$

$$u_2' = \frac{e^x}{e^x} = 1$$

$$u_2 = \int dx = x$$

$$y_p = \ln \cos x (e^x \cos x) + x (e^x \sin x)$$

$$y(x) = e^x (c_1 \cos x + c_2 \sin x) + e^x \cos x \ln \cos x + x e^x \sin x$$

2.6 Ecuaciones de Cauchy-Euler

$$x^2 y'' - 2y = 0$$

Suponemos una solución de la forma $y = x^m$.

$$y = x^m$$

$$y' = m x^{m-1}$$

$$y'' = (m-1) m x^{m-2}$$

Sustituimos en la ecuación original.

$$x^2 [(m-1)m x^{m-2}] - 2(x^m) = 0$$

$$x^2[(m-1)mx^m x^{-2}] - 2(x^m) = 0$$

$$(m-1)mx^m - 2x^m = 0$$

$$x^m[(m-1)m - 2] = 0$$

$$x^m(m^2 - m - 2) = 0$$

asi obtenemos la ecuacion auxiliar

$$m^2 - m - 2 = 0$$

$$(m+1)(m-2) = 0$$

$$m_1 = -1 ; m_2 = 2$$

Son raíces reales y distintas, asi que la solución es:

$$y = c_1 x^{-1} + c_2 x^2$$

2.

$$x^2 y'' + y = 0$$

Suponemos la solucion $y = x^m$

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Sustituimos en la ecuación

$$x^2[(m-1)mx^{m-2}] + x^m = 0$$

$$(m-1)mx^2 x^{m-2} + x^m = 0$$

$$(m^2 - m)x^m + x^m = 0$$

$$x^m(m^2 - m + 1) = 0$$

Ecuación auxiliar

$$m^2 - m + 1 = 0$$

$$m_{1,2} = \frac{1}{2} \pm \frac{1}{2}\sqrt{3}i$$

donde: $\alpha = \frac{1}{2}$ $\beta = \frac{1}{2}\sqrt{3}$

$$y = c_1 x^{\frac{1}{2} + \frac{1}{2}\sqrt{3}i} + c_2 x^{\frac{1}{2} - \frac{1}{2}\sqrt{3}i}$$

Usando la identidad,

$$x^{i\beta} = (e^{\ln x})^{i\beta} = e^{i\beta \ln x}$$

con la formula de Euler, es lo mismo que

$$x^{i\beta} = \cos(\beta \ln x) + i \operatorname{sen}(\beta \ln x)$$

$$x^{-i\beta} = \cos(\beta \ln x) - i \operatorname{sen}(\beta \ln x)$$

entonces

$$x^{i\beta} + x^{-i\beta} = \cos(\beta \ln x) + i \operatorname{sen}(\beta \ln x) + \cos(\beta \ln x) - i \operatorname{sen}(\beta \ln x) = 2\cos(\beta \ln x)$$

$$x^{i\beta} - x^{-i\beta} = \cos(\beta \ln x) + i \operatorname{sen}(\beta \ln x) - \cos(\beta \ln x) + i \operatorname{sen}(\beta \ln x) = 2i \operatorname{sen}(\beta \ln x)$$

$$\text{si } y = C_1 x^{\alpha+i\beta} + C_2 x^{\alpha-i\beta}$$

$$y_1 = x^\alpha (x^{i\beta} + x^{-i\beta}) = 2x^\alpha \cos(\beta \ln x)$$

$$y_2 = x^\alpha (x^{i\beta} - x^{-i\beta}) = 2x^\alpha i \operatorname{sen}(\beta \ln x)$$

se concluye que

$$y_1 = x^\alpha \cos(\beta \ln x) \text{ y } y_2 = x^\alpha \operatorname{sen}(\beta \ln x)$$

Así la solución general es

$$y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \operatorname{sen}(\beta \ln x)]$$

$$y = x^{\frac{1}{2}} [c_1 \cos(\frac{1}{2}\sqrt{3}\ln x) + c_2 \operatorname{sen}(\frac{1}{2}\sqrt{3}\ln x)]$$

3.

$$x^2 y'' + x y' + 4y = 0$$

Suponemos la solución:

$$y = x^m$$

$$y' = m x^{m-1}$$

$$y'' = (m-1)m x^{m-2}$$

Sustituimos en la ecuación.

$$x^2[(m-1)m x^{m-2}] + x(m x^{m-1}) + 4(x^m) = 0$$

$$x^m(m^2 - m + m + 4) = 0$$

$$x^m(m^2 + 4) = 0$$

$$m^2 = -4$$

$$m_{1,2} = \pm\sqrt{-4}$$

$$m_{1,2} = \pm 2i$$

$$\alpha = 0 \quad \beta = 2$$

$$y = x^0(c_1 \cos 2 \ln x + c_2 \sin 2 \ln x)$$

$$y = c_1 \cos 2 \ln x + c_2 \sin 2 \ln x$$

4.

$$x^2 y'' - 3xy' - 2y = 0$$

Solucion propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Sustituimos.

$$x^2[(m-1)mx^{m-2}] - 3x(mx^{m-1}) - 2(x^m) = 0$$

$$x^m[(m^2 - m) - 3m - 2] = 0$$

$$x^m(m^2 - 4m - 2) = 0$$

$$m_{1,2} = 2 \pm \sqrt{6}$$

$$y = c_2 x^{2+\sqrt{6}} + c_1 x^{2-\sqrt{6}}$$

5.

$$25x^2 y'' + 25xy' + y = 0$$

Solución propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Sustituimos.

$$25x^2[(m-1)mx^{m-2}] + 25x(mx^{m-1}) + x^m = 0$$

$$x^m[25m^2 - 25m + 25m + 1] = 0$$

$$25m^2 + 1 = 0$$

$$m_{1,2} = \pm \sqrt{-\frac{1}{25}} = \pm \frac{1}{5}i$$

$$\alpha = 0, \beta = \frac{1}{5}$$

$$y = x^0(c_1 \cos \frac{1}{5} \ln x + c_2 \operatorname{sen} \frac{1}{5} \ln x)$$

$$y = c_1 \cos \frac{1}{5} \ln x + c_2 \operatorname{sen} \frac{1}{5} \ln x$$

6.

$$x^2 y'' + 5xy' + y = 0$$

Solucion propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Sustituimos.

$$x^2[(m-1)mx^{m-2}] + 5x(mx^{m-1}) + x^m = 0$$

$$x^m(m^2 - m + 5m + 1) = 0$$

$$m^2 + 4m + 1 = 0$$

$$m_{1,2} = 2 \pm \sqrt{3}$$

$$y = c_1 x^{2+\sqrt{3}} + c_2 x^{2-\sqrt{3}}$$

7.

$$xy'' - 4y' = x^4$$

Solución propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Hacemos la ecuacion de la forma de Cauchy Euler, para esto la multiplicamos por x.

$$x^2 y'' - 4xy' = x^5$$

Resolvemos la parte homogenea.

$$y_h = x^2 y'' - 4xy' = 0$$

Sustituimos

$$x^2[(m-1)mx^{m-2}] - 4x(mx^{m-1}) = 0$$

$$x^m(m^2 - m - 4m) = 0$$

$$m(m - 5) = 0$$

$$m_1 = 0 \quad m_2 = 5$$

$$y_h = c_1 x^0 + c_2 x^5$$

$$y_h = c_1 + c_2 x^5$$

Resolvemos por variación de parámetros.

Para esto tenemos que escribir la ecuación en la forma estándar $P(x)y'' +$

$$Q(x)y' = f(x)$$

Dividimos la ecuación original entre x

$$y'' - 4\frac{y}{x} = x^3$$

identificamos $f(x) = x^3$

Definimos y_1, y_2

$$y_1 = 1, \quad y'_1 = 0$$

$$y_2 = x^5, \quad y'_2 = 5x^4$$

$$W = \begin{vmatrix} 1 & x^5 \\ 0 & 5x^4 \end{vmatrix} = 5x^4 - 0 = 5x^4$$

$$W_1 = \begin{vmatrix} 0 & x^5 \\ x^3 & 5x^4 \end{vmatrix} = 0 - x^8 = -x^8$$

$$W_2 = \begin{vmatrix} 1 & 0 \\ 0 & x^3 \end{vmatrix} = x^3$$

Calculamos u_1, u_2

$$u'_1 = \frac{-x^8}{5x^4} = -\frac{1}{5}x^4$$

$$u_1 = -\frac{1}{5} \int x^4 dx = -\frac{1}{25}x^5$$

$$u'_2 = \frac{x^3}{5x^4} = \frac{1}{5x}$$

$$u_2 = \frac{1}{5} \int \frac{1}{x} dx = \frac{1}{5} \ln x$$

$$y_p = -\frac{1}{25}x^5(1) + \frac{1}{5} \ln x(x^5)$$

$$y_p = -\frac{1}{25}x^5 + \frac{x^5}{5} \ln x$$

$$y(x) = y_h + y_p$$

$$y(x) = c + c_2 x^5 - \frac{1}{25}x^5 + \frac{x^5}{5} \ln x$$

7.

$$x^2 y'' - xy' + y = 2x$$

Solución propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

Resolvemos la ecuación homogénea asociada.

$$y_h = x^2 y'' - xy' + y = 0$$

Sustituimos en la ecuación original.

$$x^2[(m-1)mx^{m-2}] - x(mx^{m-1}) + x^m = 0$$

$$m^2 - m - m + 1 = 0$$

$$m^2 - 2m + 1 = 0$$

$$(m-1)^2$$

$$m_{1,2} = 1$$

$$y_h = c_1 x + c_2 x \ln x$$

Ponemos la ecuación en la forma estandar

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = \frac{2}{x}$$

$$\text{Identificamos } f(x) = \frac{2}{x}$$

$$\text{Identificamos } y_1 = x, y_2 = x \ln x \text{ y } y'_1 = 1, y'_2 = \ln x + 1$$

Calculamos los Wronskianos

$$W = \begin{vmatrix} x & x \ln x \\ 1 & \ln x + 1 \end{vmatrix} = (x)(\ln x + 1) - (x \ln x)(1) = x \ln x + x - x \ln x = x$$

$$x \ln x + x = x \ln \frac{x}{x} + x = x \ln(1) + x = x$$

$$W_1 = \begin{vmatrix} 0 & x \ln x \\ \frac{2}{x} & \ln x + 1 \end{vmatrix} = (x \ln x)(\frac{2}{x}) - \frac{2}{x} x \ln x = 2 \ln x - 2 \ln x = 0$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & \frac{2}{x} \end{vmatrix} = \frac{2}{x} x - 0 = 2$$

Calculamos u_1, u_2

$$u'_1 = \frac{\frac{2}{x} x \ln x}{x} = \frac{2 \ln x}{x^2}$$

$$u_1 = 2 \int \frac{\ln x}{x^2} = -\frac{\ln x + 1}{x}$$

$$u'_2 = \frac{2}{x}$$

$$u_2 = 2 \int \frac{1}{x} = 2 \ln x$$

$$y_p = y_1 u_1 + y_2 u_2 = x \left(-\frac{\ln x + 1}{x} \right) + x \ln x (2 \ln x) = -\ln x + 1 + 2x \ln^2 x$$

8.

$$x^2 y'' - 2xy' + 2y = x^4 e^x$$

Solución propuesta.

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = (m-1)mx^{m-2}$$

$$x^2[(m-1)mx^{m-2}] - 2x(mx^{m-1}) + 2x^m = x^4 e^x$$

Solucionamos la ecuacion homogenea

$$x^2 y'' - 2xy' + 2y = 0$$

$$x^m(m^2 - m - 2m + 2) = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m-2)(m-1) = 0$$

$$m_1 = 2, m_2 = 1$$

$$y_h = c_1 x^2 + c_2 x$$

Convertimos la ecuacion original a la forma estandar

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = x^2 e^x$$

$$\text{Definimos } y_1, y_2, f(x) = x^2 e^x$$

$$y_1 = x^2; y'_1 = 2x$$

$$y_2 = x; y'_2 = 1$$

Calculamos el Wronskiano

$$W = \begin{vmatrix} x^2 & x \\ 2x & 1 \end{vmatrix} = x^2 - 2x^2 = -x^2$$

$$W_1 = \begin{vmatrix} 0 & x \\ x^2 e^x & 1 \end{vmatrix} = 0 - x^3 e^x = -x^3 e^x$$

$$W_2 = \begin{vmatrix} x^2 & 0 \\ 2x & x^2 e^x \end{vmatrix} = x^4 e^x$$

Calculamos u_1, u_2

$$u'_1 = \frac{-x^3 e^x}{-x^2} = x e^x$$

$$u_1 = \int x e^x dx = e^x (x - 1)$$

$$u'_2 = \frac{x^4 e^x}{-x^2} = -x^2 e^x$$

$$u'_2 = -\int x^2 e^x dx = e^x (x^2 - 2x + 2)$$

$$y_p = u_1 y_1 + u_2 y_2 = [e^x (x - 1)]x^2 + [e^x (x^2 - 2x + 2)]x$$

$$y_p = x^2 e^x (x - 1) + x e^x (x^2 - 2x + 2)$$

$$y(x) = y_p + y_h$$

$$y(x) = c_1 x^2 + c_2 x + x^2 e^x (x - 1) + x e^x (x^2 - 2x + 2)$$

9.

3 Soluciones en series de potencias

1.

$$y'' - xy = 0$$

Substituyendo $y = \sum_{n=0}^{\infty} c_n x^n$ y la segunda derivada $y'' = \sum_{n=2}^{\infty} (n-1)nc_n x^{n-2}$

$$\sum_{n=2}^{\infty} (n-1)nc_n x^{n-2} - x \left(\sum_{n=0}^{\infty} c_n x^n \right) = 0$$

$$\sum_{n=2}^{\infty} (n-1)nc_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Ahora sumamos las dos series igualando los indices de ambas sumas.

$$2(1)c_2 x^0 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

$$2c_2 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

hacemos $k = n - 2$ para la primera serie y $k = n + 1$ para la segunda serie, de modo que: $n = k + 2$, $n = k - 1$ respectivamente.

Substituimos

$$2c_2 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

$$2c_2 \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$$

$$2c_2 \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}] x^k = 0$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

$$c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$$

Esta relacion genera coeficientes consecutivos de la solucion propuesta, con el valor de k como enteros positivos.

Ahora

$$2c_2 = 0 ; c_2 = 0$$

$$k = 1 , c_3 = \frac{c_0}{3(2)} = \frac{1}{6}c_0$$

$$k = 2 , c_4 = \frac{c_1}{4(3)} = \frac{1}{12}c_1$$

$$k = 3 , c_5 = \frac{c_2}{5(4)} = \frac{1}{20}c_2 = 0 \leftarrow c_2 = 0$$

$$k = 4 , c_6 = \frac{c_3}{6(5)} = \frac{1}{30}(\frac{1}{6})c_0 = \frac{1}{180}c_0$$

$$k = 5 , c_7 = \frac{c_4}{7(6)} = \frac{1}{42}(\frac{1}{12})c_1 = \frac{1}{504}c_1$$

$$k = 6 , c_8 = \frac{c_5}{8(7)} = 0 \leftarrow c_5 = 0$$

$$k = 7 , c_9 = \frac{c_6}{9(8)} = \frac{1}{72}(\frac{1}{180})c_0 = \frac{1}{12960}c_0$$

$$k = 8 , c_{10} = \frac{c_7}{10(9)} = \frac{1}{10(9)(504)}c_1$$

$$k = 9 , c_{11} = \frac{c_8}{11(10)} = 0 \leftarrow c_8 = 0$$

Sustituyendo coeficientes en la suposicion original

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + c_{11}x^{11} + \dots,$$

$$y = c_0 + c_1x + 0 + \frac{1}{6}c_0x^3 + \frac{1}{12}c_1x^4 + 0 + \frac{1}{180}c_0x^6 + \frac{1}{504}c_1x^7 + 0 + \frac{1}{12960}c_0x^9 + \frac{1}{90(504)}c_1x^{10} + 0$$

$$y = c_0(1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \frac{1}{12960}x^9) + c_1(x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \frac{1}{90(504)}x^{10})$$

2

$$y''' + x^2y' + xy = 0$$

Sutituyendo:

$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} (n-1)n c_n x^{n-2}$$

En la ecuación original

$$\sum_{n=2}^{\infty} (n-1)n c_n x^{n-2} + x^2 [\sum_{n=1}^{\infty} c_n n x^{n-1}] + x [\sum c_n x^n] = 0$$

$$\sum_{n=2}^{\infty} (n-1)nc_nx^{n-2} + \sum_{n=1}^{\infty} c_nnx^{n+1} + \sum_{n=0}^{\infty} c_nx^{n+1} = 0$$

$$2c_2x^0 + 6c_3x \sum_{n=4}^{\infty} (n-1)nc_nx^{n-2} + \sum_{n=1}^{\infty} c_nnx^{n+1} + c_0x^1 \sum_{n=1}^{\infty} c_nx^{n+1} = 0$$

Hacemos $k = n-2$ para la primera serie, y $k = n+1$ para la segunda y tercera series.

$$2c_2x^0 + 6c_3x \sum_{k=2}^{\infty} (k+2-1)(k+2)c_{k+2}x^{k+2-2} + \sum_{k=2}^{\infty} c_{k-1}(k-1)x^{k-1+1} + c_0x^1 \sum_{k=2}^{\infty} c_{k-1}x^{k-1+1} = 0$$

$$2c_2 + 6c_3x + c_0x \sum_{k=2}^{\infty} (k+1)(k+2)c_{k+2}x^k + c_{k-1}(k-1)x^k + c_{k-1}x^k = 0$$

$$2c_2 + 6c_3x + c_0x \sum_{k=2}^{\infty} [(k+1)(k+2)c_{k+2} + c_{k-1}(k-1) + c_{k-1}]x^k = 0$$

$$(k+1)(k+2)c_{k+2} + c_{k-1}(k-1) + c_{k-1}$$

Entonces tenemos

$$2c_2 = 0 ; c_2 = 0$$

$$6c_3 + c_0 = 0$$

$$c_3 = -\frac{1}{6}c_0$$

$$c_{k+2} = \frac{[(k-1)+1]c_{k-1}}{(k+1)(k+2)} = \frac{kc_{k-1}}{(k+1)(k+2)}$$

Sustituyendo $k = 2, 3, 4, \dots$ en la formula se obtiene

$$c_4 = \frac{2c_1}{3(4)} = \frac{1}{6}c_1$$

$$c_5 = \frac{3c_2}{4(5)} = 0 \leftarrow c_2 = 0$$

$$c_6 = \frac{4c_3}{5(6)} = \frac{2}{15}(-\frac{1}{6}c_0) = -\frac{1}{45}c_0$$

$$c_7 = \frac{5c_4}{6(7)} = \frac{5}{42}(\frac{1}{6}c_1) = \frac{5}{136}c_1$$

$$c_8 = \frac{6c_5}{7(8)} = \frac{6}{56}(0) = 0$$

$$c_9 = \frac{7c_6}{8(9)} = \frac{7}{72}(-\frac{1}{45}c_0) = -\frac{7}{72(45)}c_0$$

$$c_{10} = \frac{8c_7}{9(10)} = \frac{4}{45}(\frac{5}{136}c_1) = \frac{5}{45(34)}c_1$$

$$c_{11} = \frac{9c_8}{10(11)} = \frac{9}{110}(0) = 0$$

$$c_{12} = \frac{10c_9}{11(12)} = \frac{5}{66}(-\frac{7}{72(45)}c_0) = -\frac{7}{66(72)(9)}c_0$$

Por tanto,

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + \dots$$

$$y = c_1[\frac{1}{6}x^4 + \frac{5}{136}x^7 + \frac{5}{9(34)}x^{10}] - c_0[\frac{1}{45}x^6 + \frac{7}{72(45)}x^9 + \frac{7}{66(72)(9)}x^{12}]$$

3.

$$y'' - 2xy' + y = 0$$

Sutituyendo:

$$y = \sum_{n=0}^{\infty} c_nx^n$$

$$y' = \sum_{n=1}^{\infty} c_nnx^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} (n-1)nc_nx^{n-2}$$

En la ecuación original

$$\sum_{n=2}^{\infty} (n-1)nc_nx^{n-2} - 2x [\sum_{n=1}^{\infty} c_nnx^{n-1}] + \sum_{n=0}^{\infty} c_nx^n = 0$$

$$\sum_{n=2}^{\infty} (n-1)nc_nx^{n-2} - 2 \sum_{n=1}^{\infty} c_nnx^n + \sum_{n=0}^{\infty} c_nx^n = 0$$

$$2c_2 \sum_{n=3}^{\infty} (n-1)nc_nx^{n-2} - 2 \sum_{n=1}^{\infty} c_nnx^n + c_0 \sum_{n=1}^{\infty} c_nx^n = 0$$

Hacemos $k = n-2$ para la serie uno y $k = n$ para las dos y tres.

$$2c_2 \sum_{k=1}^{\infty} (k+2-1)(k+2)c_{k+2}x^{k+2-2} - 2 \sum_{k=1}^{\infty} c_kkx^k + c_0 \sum_{k=1}^{\infty} c_kx^k = 0$$

$$2c_2 + c_0 \sum_{k=1}^{\infty} (k+1)(k+2)c_{k+2}x^k + \sum_{k=1}^{\infty} c_kkx^k + \sum_{k=1}^{\infty} c_kx^k = 0$$

$$2c_2 + c_0 \sum_{k=1}^{\infty} (k+1)(k+2)c_{k+2}x^k - 2c_kkx^k + c_kx^k = 0$$

$$2c_2 + c_0 \sum_{k=1}^{\infty} (k+1)(k+2)c_{k+2}x^k - 2c_kkx^k + c_kx^k = 0$$

De esta igualdad se concluye que

$$2c_2 + c_0 = 0$$

$$c_2 = -\frac{1}{2}c_0$$

$$(k+1)(k+2)c_{k+2}x^k - 2c_k kx^k + c_k x^k = 0$$

$$[(k+1)(k+2)c_{k+2} - 2c_k k + c_k]x^k = 0$$

$$(k+1)(k+2)c_{k+2} - 2c_k k + c_k = 0$$

$$c_{k+2} = \frac{(2k+1)c_k}{(k+1)(k+2)}$$

Sustituyendo $k = 1, 2, 3, 4, \dots$

$$c_3 = \frac{3c_1}{2(3)} = \frac{1}{2}c_1$$

$$c_4 = \frac{5c_2}{3(4)} = \frac{5}{12}\left(-\frac{1}{2}c_0\right) = -\frac{5}{24}c_0$$

$$c_5 = \frac{7c_3}{4(5)} = \frac{7}{20}\left(\frac{1}{2}c_1\right) = \frac{7}{40}c_1$$

$$c_6 = \frac{9c_4}{5(6)} = \frac{9}{30}\left(-\frac{5}{24}c_0\right) = -\frac{1}{16}c_0$$

$$c_7 = \frac{11c_5}{6(7)} = \frac{11}{42}\left(\frac{7}{40}c_1\right) = \frac{11}{6(40)}c_1$$

$$c_8 = \frac{13c_6}{7(8)} = \frac{13}{56}\left(-\frac{1}{16}c_0\right) = -\frac{13}{56(16)}c_0$$

$$c_9 = \frac{15c_7}{8(9)} = \frac{15}{72}\left(\frac{11}{6(40)}c_1\right) = \frac{161}{72(240)}c_1$$

$$c_{10} = \frac{17c_8}{9(10)} = -\frac{17(13)}{9(10)(56)(16)}c_0$$

$$c_{11} = \frac{18c_9}{10(11)} = \frac{1}{55}\left(\frac{161}{8(240)}\right)c_1$$

$$y = c_1 \left[\frac{1}{2}x^3 + \frac{7}{40}x^5 + \frac{11}{240}x^7 + \frac{161}{72(240)}x^9 + \frac{161}{55(8)(240)}x^{11} \right] -$$

$$c_0 \left[\frac{5}{24}x^4 + \frac{1}{16}x^6 + \frac{13}{56(16)}x^8 + \frac{17(13)}{90(56)(16)}x^{10} \right]$$

4.

$$(x^2 + 2)y'' + 3xy' - y = 0$$

Sutituyendo:

$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} (n-1)c_n x^{n-2}$$

$$(x^2 + 2) \sum_{n=2}^{\infty} (n-1)c_n x^{n-2} + 3x \sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

$$x^2 \sum_{n=2}^{\infty} (n-1)c_n x^{n-2} + 2 \sum_{n=2}^{\infty} (n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} 3c_n n x^n - \sum_{n=0}^{\infty} c_n x^n =$$

0

$$\sum_{n=2}^{\infty} (n-1)nc_nx^n + \sum_{n=2}^{\infty} 2(n-1)nc_nx^{n-2} + \sum_{n=1}^{\infty} 3c_nnx^n - \sum_{n=0}^{\infty} c_nx^n = 0$$

$$\sum_{n=2}^{\infty} (n-1)nc_nx^n + \sum_{n=2}^{\infty} 2(n-1)nc_nx^{n-2} + 3c_1x \sum_{n=2}^{\infty} 3c_nnx^n - c_0 + c_1x \sum_{n=2}^{\infty} c_nx^n = 0$$

$$3c_1x + c_0 + c_1x \sum_{n=2}^{\infty} (n-1)nc_nx^n + 2(2-1)2c_2x^{2-2} + 2(3-1)3c_3x^{3-2} \sum_{n=4}^{\infty} 2(n-1)nc_nx^{n-2} + \sum_{n=2}^{\infty} 3c_nnx^n - \sum_{n=2}^{\infty} c_nx^n = 0$$

$$3c_1x + c_0 + c_1x + 4c_2 + 12c_3x \sum_{n=2}^{\infty} (n-1)nc_nx^n + \sum_{n=4}^{\infty} 2(n-1)nc_nx^{n-2} + \sum_{n=2}^{\infty} 3c_nnx^n - \sum_{n=2}^{\infty} c_nx^n = 0$$

Hacemos $k = n - 2$ para la segunda serie y $k = n$ para las demas

$$3c_1x + c_0 + c_1x + 4c_2 + 12c_3x \sum_{k=2}^{\infty} (k-1)kc_kx^k + \sum_{k=2}^{\infty} 2(k+2-1)(k+2)c_{k+2}x^{k+2-2} + \sum_{k=2}^{\infty} 3c_kkx^k - \sum_{k=2}^{\infty} c_kx^k = 0$$

$$4c_1x + c_0 + 4c_2 + 12c_3x \sum_{k=2}^{\infty} [(k-1)kc_k + 2(k+1)(k+2)c_{k+2} + 3c_kk - c_k]x^k = 0$$

De esta igualdad se obtiene

$$4c_1 + 12c_3 = 0$$

$$c_3 = -\frac{c_1}{3}$$

$$c_0 + 4c_2 = 0$$

$$c_2 = -\frac{c_0}{4}$$

$$(k-1)kc_k + 2(k+1)(k+2)c_{k+2} + 3c_kk - c_k$$

$$c_{k+2} = \frac{-3kc_k + c_k - (k-1)kc_k}{2(k+1)(k+2)} = \frac{[-3k+1-k^2-k]c_k}{2(k+1)(k+2)} = \frac{[-4k+1-k^2]c_k}{2(k+1)(k+2)}$$

Sustituyendo valores de $k = 2, 3, 4, 5, \dots$

$$c_2 = -\frac{c_0}{4}$$

$$c_3 = -\frac{c_1}{3}$$

$$c_4 = \frac{(-6+1-4-2)c_2}{2(3)(4)} = -\frac{11}{24}(-\frac{1}{4}c_0) = \frac{11}{96}c_0$$

$$c_5 = \frac{(-12+1-9)c_3}{2(4)(5)} = -\frac{20}{40}c_3 = -\frac{1}{2}(-\frac{1}{3}c_1) = \frac{1}{6}c_1$$

$$c_6 = \frac{(-16+1-16)c_4}{2(5)(6)} = -\frac{31}{60}c_4 = -\frac{31}{60}(\frac{11}{96}c_0) = \frac{31(11)}{(60)(96)}c_0$$

$$c_7 = \frac{(-20+1-25)c_5}{2(6)(7)} = \frac{-44}{84}c_5 = \frac{11}{21}(\frac{1}{6}c_1) = \frac{11}{126}c_1$$

$$c_8 = \frac{(-24+1-36)c_6}{2(7)(8)} = \frac{59}{112}(\frac{11(31)}{60(96)}c_0) = \frac{11(31)}{112(60)(96)}c_0$$

$$y = c_0[-\frac{1}{4}x^2 + \frac{11}{90}x^4 + \frac{31(11)}{60(96)}x^6 + \frac{11(31)}{112(60)(96)}x^8] + c_1[-\frac{1}{3}x^3 + \frac{1}{6}x^5 + \frac{11}{126}x^7]$$

4 Transformada de Laplace

1.

$$f(t) \begin{cases} -1, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st}f(t)dt = -\int_0^1 e^{-st}(1)dt + \int_1^{\infty} e^{-st}(1)dt$$

$$= -\frac{e^{-st}}{-s} \Big|_0^1 + \frac{e^{-st}}{-s} \Big|_1^{\infty}$$

$$\begin{aligned}
&= -\frac{e^{-s(1)}}{-s} - \left[-\frac{e^{-s(0)}}{-s}\right] + \frac{e^{-s(1)}}{-s} - \frac{e^{-s(\infty)}}{-s} \\
&= \frac{e^{-s}}{s} - \frac{1}{s} + \frac{e^{-s}}{s} + \frac{0}{s} \\
&= \frac{2e^{-s}}{s} - \frac{1}{s}
\end{aligned}$$

2.

$$f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

$$\begin{aligned}
\mathcal{L}[f(t)] &= \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} t dt + \int_1^\infty e^{-st} (1) dt \\
&= -\frac{e^{-st}}{s} \left(t - \frac{1}{s}\right) \Big|_0^1 + -\frac{e^{-st}}{s} \Big|_1^\infty \\
&= -\frac{e^{-s(1)}}{s} \left(1 - \frac{1}{s}\right) - \left[-\frac{e^{-s(0)}}{s} \left(0 - \frac{1}{s}\right)\right] + \left[-\frac{e^{-s(\infty)}}{s} - \frac{e^{-s(0)}}{s}\right] \\
&= -\frac{e^{-s}}{s} + \frac{e^{-s}}{s^2} + \frac{1}{s^2} - \frac{1}{s}
\end{aligned}$$

$$f(t) = te^{4t}$$

$$\begin{aligned}
\mathcal{L}\{te^{4t}\} &= \int_0^\infty e^{-st} te^{4t} dt = \int_0^\infty te^{-(s-4)t} dt \\
&= -\frac{e^{-(s-4)t}}{(s-4)^2} [-s+3] \Big|_0^\infty \\
&= -\frac{e^{-(s-4)\infty}}{(s-4)^2} - \left[-\frac{e^{-(s-4)0}}{(s-4)^2}\right] \\
&= \frac{0}{(s-4)^2} + \frac{1}{(s-4)^2} \\
&= \frac{1}{(s-4)^2}
\end{aligned}$$

3.

$$y'' + 3y' + 2y = 0$$

$$y(0) = 1, y'(0) = 1$$

Aplicamos transformada de Laplace a toda la ecuación

$$\mathcal{L}[y''] + 3\mathcal{L}[y'] + 2\mathcal{L}[y] = 0$$

$$[s^2 Y(s) - sy(0) - y'(0)] + 3[sY(s) - y(0)] + 2[Y(s)] = 0$$

$$s^2 Y(s) - sy(0) - y'(0) + 3sY(s) - 3y(0) + 2Y(s) = 0$$

Sustituimos los valores iniciales

$$s^2 Y(s) - s(1) - 1 + 3sY(s) - 3(1) + 2Y(s) = 0$$

$$s^2 Y(s) - s - 1 + 3sY(s) - 3 + 2Y(s) = 0$$

Factorizando

$$Y(s)(s^2 + 3s + 2) - s - 4$$

$$Y(s) = \frac{4+s}{s^2+3s+2}$$

Separamos en fracciones parciales

$$\frac{4+s}{(s+2)(s+1)} = \frac{A}{s+1} + \frac{B}{s+2}$$

Por el método de Heaviside

$$A = \frac{(4+s)(s+1)}{(s+2)(s+1)} \Big|_{s=-1} = \frac{4+(-1)}{-1+2} = 3$$

$$B = \frac{(4+s)(s+2)}{(s+1)(s+2)} \Big|_{s=-2} = \frac{4-2}{-2+1} = -2$$

Sustituimos en la ecuacion transformada.

$$Y(s) = \frac{3}{s+1} - \frac{2}{s+2}$$

Aplicamos la transformada inversa a cada término del desarrollo anterior.

$$y(t) = \mathcal{L}^{-1}\left[\frac{3}{s+1}\right] - \mathcal{L}^{-1}\left[\frac{2}{s+2}\right]$$

$$y(t) = 3\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] - 2\mathcal{L}^{-1}\left[\frac{1}{s+2}\right]$$

$$y(t) = 3e^{-t} - 2e^{-2t}$$

$$y'' - xy = 0$$

Sutituyendo $y = \sum_{n=0}^{\infty} c_n x^n$ y la segunda derivada $y'' = \sum_{n=2}^{\infty} (n-1)nc_n x^{n-2}$

$$\sum_{n=2}^{\infty} (n-1)nc_n x^{n-2} - x \left(\sum_{n=0}^{\infty} c_n x^n \right) = 0$$

$$\sum_{n=2}^{\infty} (n-1)nc_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Ahora sumamos las dos series igualando los indices de ambas sumas.

$$2(1)c_2 x^0 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

$$2c_2 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

hacemos $k = n - 2$ para la primera serie y $k = n + 1$ para la segunda serie, de modo que: $n = k + 2$, $n = k - 1$ respectivamente.

Sutituimos

$$2c_2 \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

$$2c_2 \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$$

$$2c_2 \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}] x^k = 0$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

$$c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$$

Esta relacion genera coeficientes consecutivos de la solucion propuesta, con el valor de k como enteros positivos.

Ahora

$$2c_2 = 0 ; c_2 = 0$$

$$k = 1 , c_3 = \frac{c_0}{3(2)} = \frac{1}{6}c_0$$

$$k = 2 , c_4 = \frac{c_1}{4(3)} = \frac{1}{12}c_1$$

$$k = 3 , c_5 = \frac{c_2}{5(4)} = \frac{1}{20}c_2 = 0 \leftarrow c_2 = 0$$

$$k = 4 , c_6 = \frac{c_3}{6(5)} = \frac{1}{30}(\frac{1}{6})c_0 = \frac{1}{180}c_0$$

$$k = 5 , c_7 = \frac{c_4}{7(6)} = \frac{1}{42}(\frac{1}{12})c_1 = \frac{1}{504}c_1$$

$$k = 6 , c_8 = \frac{c_5}{8(7)} = 0 \leftarrow c_5 = 0$$

$$k = 7 , c_9 = \frac{c_6}{9(8)} = \frac{1}{72}(\frac{1}{180})c_0 = \frac{1}{12960}c_0$$

$$k = 8 , c_{10} = \frac{c_7}{10(9)} = \frac{1}{10(9)(504)}c_1$$

$$k = 9 , c_{11} = \frac{c_8}{11(10)} = 0 \leftarrow c_8 = 0$$

Sustituyendo coeficientes en la suposicion original

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + c_{11}x^{11} + \dots,$$

$$y = c_0 + c_1x + 0 + \frac{1}{6}c_0x^3 + \frac{1}{12}c_1x^4 + 0 + \frac{1}{180}c_0x^6 + \frac{1}{504}c_1x^7 + 0 + \frac{1}{12960}c_0x^9 + \frac{1}{90(504)}c_1x^{10} + 0$$

$$y = c_0(1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \frac{1}{12960}x^9) + c_1(x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \frac{1}{90(504)}x^{10})$$

2.

$$y'' - (x+1)y' - y = 0$$

Sutituyendo $y = \sum_{n=0}^{\infty} c_n x^n$ la primera derivada $\sum_{n=1}^{\infty} c_n n x^{n-1}$ y la segunda derivada $y'' = \sum_{n=2}^{\infty} (n-1) n c_n x^{n-2}$

$$\sum_{n=2}^{\infty} (n-1) n c_n x^{n-2} - (x+1) \sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n-1) n c_n x^{n-2} - \sum_{n=1}^{\infty} c_n n x^n - \sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

$$2c_2x^0 \sum_{n=3}^{\infty} (n-1)nc_nx^{n-2} - \sum_{n=1}^{\infty} c_nnx^n - c_1x^0 \sum_{n=2}^{\infty} c_nnx^{n-1} - c_0x^0 \sum_{n=1}^{\infty} c_nx^n = 0$$

hacemos $k = n - 2$ para la primera serie, $k = n - 1$ para la tercera, $k = n$ para la segunda y la cuarta serie.

$$c_2 - c_1 - c_0 + \sum_{n=3}^{\infty} (n-1)nc_nx^{n-2} - \sum_{n=1}^{\infty} c_nnx^n - \sum_{n=2}^{\infty} c_nnx^{n-1} - \sum_{n=1}^{\infty} c_nx^n = 0$$

$$c_2 - c_1 - c_0 + \sum_{k=1}^{\infty} (k+2-1)(k+2)c_{k+2}x^k - \sum_{k=1}^{\infty} c_kkx^k - \sum_{k=1}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=1}^{\infty} c_kx^k = 0$$

$$c_2 - c_1 - c_0 + \sum_{k=1}^{\infty} [(k+1)(k+2)c_{k+2} - kc_k - (k+1)c_{k+1} - c_k]x^k = 0$$

De aki se concluye que

$$c_2 = 0$$

$$c_1 = 0$$

$$c_0 = 0$$

$$(k+1)(k+2)c_{k+2} - (k+1)c_{k+1} - kc_k - c_k = 0$$

$$c_{k+2} = \frac{(k+1)c_{k+1} + (k+1)c_k}{(k+1)(k+2)}$$

Sustituyendo $k = 1, 2, 3, \dots$,

$$k = 1, c_3 = \frac{2c_2 + 2c_1}{2(3)} = 0$$

$$k = 2, c_4 = \frac{3c_3 + 3c_2}{3(4)} = \frac{3}{12}c_3 = 0$$

$$k = 3, c_5 = \frac{4c_4 + 4c_3}{4(5)} = 0$$

Serie de Taylor

1.

$$y' = x + 2y^2$$

$$y(0) = 0$$

$$y'(0) = 1$$

Derivando

$$y'' = 1 + 4yy'$$

$$y''' = 4y'y' + 4yy''$$

$$y^{iv} = 4y''y' + 4y'y'' + 4y'y'' + 4yy''' = 12y'y'' + 4yy'''$$

$$y^v = 12y''y'' + 12y'y''' + 4y''y''' + 4yy^{iv}$$

$$y^{vi} = 12y'''y'' + 12y''y''' + 12y''y''' + 12y'y^{iv} + 4y'''y''' + 4y''y^{iv} + 4y''y^{iv} + 4yy^v = 36y''y''' + 12y'y^{iv} + 4y'''y''' + 4y''y^{iv} + 4y''y^{iv} + 4yy^v$$

$$y''(0) = 1, y'''(0) = 4, y^{iv}(0) = 12, y^v(0) = 76, y^{vi}(0) = 408$$

$$y(x) = \frac{x}{1!} + \frac{x^2}{2!} + \frac{4x^3}{3!} + \frac{12x^4}{4!} + \frac{76x^5}{5!} + \frac{408x^6}{6!}$$

1.

$$f(t) = 4t - 10$$

$$\mathcal{L}[f(t)] = 4\mathcal{L}[t] - 10\mathcal{L}[1]$$

$$\mathcal{L}[f(t)] = 4 \int_0^\infty te^{-st}dt - 10 \int_0^\infty e^{-st}dt$$

Integramos por partes la primera integral

$$4 \int_0^\infty te^{-st}dt =$$

$$u = t, du = 1$$

$$dv = e^{-st}, v = -\frac{e^{-st}}{s}$$

$$= 4[(1)(-\frac{e^{-st}}{s})|_0^\infty + \frac{1}{s} \int_0^\infty (1)(e^{-st})dt]$$

$$= -\frac{4}{s}e^{-s(\infty)} - (-\frac{4}{s}e^{-s(0)}) + \frac{1}{s} \frac{e^{-st}}{s}|_0^\infty$$

$$= \frac{4}{s} + \frac{1}{s} \left[\left(\frac{e^{-s(\infty)}}{s} - \frac{e^{-s(0)}}{s} \right) \right]$$

$$= \frac{4}{s} + \frac{1}{s}(-\frac{1}{s}) = \frac{4}{s} - \frac{1}{s^2}$$

Hacemos la segunda integral

$$10 \int_0^\infty e^{-st}dt = -\frac{e^{-st}}{s}|_0^\infty$$

$$= -\frac{e^{-s(\infty)}}{s} - (-\frac{e^{-s(0)}}{s})$$

$$= \frac{1}{s}$$

entonces

$$\mathcal{L}[f(t)] = \frac{4}{s} - \frac{1}{s^2} - \frac{1}{s} = \frac{3}{s} - \frac{1}{s^2}$$

2.

$$f(t) = e^{t/5}$$

$$\mathcal{L}[f(t)] = \mathcal{L}[e^{t/5}] = \int_0^\infty e^{t/5}e^{-st}dt$$

$$= \int_0^\infty e^{\frac{t}{5}(1-5s)}dt = \frac{e^{\frac{t}{5}(1-5s)}}{\frac{1}{5}(1-5s)}|_0^\infty$$

$$= \frac{e^{\frac{\infty}{5}(1-5s)}}{\frac{1}{5}(1-5s)} - \frac{e^{\frac{0}{5}(1-5s)}}{\frac{1}{5}(1-5s)} = -\frac{5}{1-5s} = \frac{5}{5s-1}$$

3.

$$f(t) = e^{t-2}$$

$$\mathcal{L}[f(t)] = \mathcal{L}[e^{t-2}]$$

$$\begin{aligned}
&= \int_0^\infty e^{t-2} e^{-st} dt = e^{-2} \int e^{t(1-s)} dt \\
&= e^{-2} \left[\frac{e^{t(1-s)}}{1-s} \right]_0^\infty = (e^{-2}) \left(\frac{e^{(\infty)(1-s)}}{1-s} \right) - (e^{-2}) \left(\frac{e^{(0)(1-s)}}{1-s} \right) \\
&= -\frac{e^{-2}}{1-s} = \frac{e^{-2}}{s-1}
\end{aligned}$$

4.

$$f(t) = e^t \cos t$$

$$\mathcal{L}[f(t)] = \mathcal{L}[e^t \cos t]$$

Por el teorema de traslacion del eje s

$$\mathcal{L}[f(t)] = F(s)$$

$$\mathcal{L}[e^{at} f(t)] = F(s-a)$$

$$a = 1$$

$$\mathcal{L}[\cos t] = \frac{s}{s^2+1} \implies \mathcal{L}[e^t \cos t] = \frac{s}{(s-1)^2+1}$$

5.

$$f(t) = e^{-t} \cos t$$

$$\mathcal{L}[f(t)] = \mathcal{L}[e^{-t} \cos t]$$

Por el teorema de traslacion del eje s

$$\mathcal{L}[f(t)] = F(s)$$

$$\mathcal{L}[e^{at} f(t)] = F(s-a)$$

$$a = -1$$

$$\mathcal{L}[\cos t] = \int_0^\infty e^{-st} (\cos t) dt$$

Integramos por partes

$$u = \cos t, \quad du = -\sin t$$

$$dv = e^{-st}, \quad v = \int e^{-st} dt = -\frac{e^{-st}}{s}$$

$$\int_0^\infty e^{-st} (\cos t) dt = \cos t \left(-\frac{e^{-st}}{s} \right) \Big|_0^\infty - \int_0^\infty \frac{-e^{-st}}{s} (-\sin t) dt$$

$$= -(\cos t) \frac{e^{-st}}{s} \Big|_0^\infty - \frac{1}{s} \int_0^\infty e^{-st} (\sin t) dt$$

Evaluamos el primer término y volvemos a integrar por partes el segundo término

$$u = \sin t, \quad du = \cos t$$

$$dv = e^{-st}, \quad v = -\frac{e^{-st}}{s}$$

$$= -\left[(\cos \infty) \frac{e^{-\infty}}{s} - (\cos 0) \frac{e^{-0}}{s} \right] - \frac{1}{s} \left[\sin t \left(-\frac{e^{-st}}{s} \right) \Big|_0^\infty - \int_0^\infty \left(-\frac{e^{-st}}{s} \right) \cos t dt \right]$$

$$= \left(1 + \frac{1}{s^2} \right) \left(\int_0^\infty e^{-st} \cos t dt \right) = \frac{1}{s^2 e^{st}} \sin t \Big|_0^\infty - \frac{1}{s e^{st}} \cos t \Big|_0^\infty = \frac{1}{s}$$

$$\therefore \int_0^\infty e^{-st} \cos t dt = \frac{\frac{1}{s}}{1 + \frac{1}{s^2}} = \frac{s}{s^2 + 1}$$

Aplicando el teorema de traslacion del eje s

$$\mathcal{L}[f(t)] = F(s)$$

$$\mathcal{L}[e^{at} f(t)] = F(s-a)$$

$$a = -1$$

$$\mathcal{L}[e^{-t} \cos t] = \frac{s+1}{(s+1)^2 + 1}$$

6.

$$f(t) = \begin{cases} -1 & 0 < t < 2 \\ 0 & 2 \leq t < 4 \\ 1 & t \geq 4 \end{cases}$$

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t) dt = \int_0^2 e^{-st} (-1) dt + \int_2^4 e^{-st} (0) dt + \int_4^\infty e^{-st} (1) dt$$

$$\int_0^2 e^{-st} (-1) dt = -\left[-\frac{1}{s} e^{-st}\right]_0^2 = \frac{1}{s} e^{-2s} - \frac{1}{s} e^0 = \frac{1}{s} [e^{-2s} - 1]$$

$$\int_2^4 e^{-st} (0) dt = 0$$

$$\int_4^\infty e^{-st} (1) dt = -\frac{1}{s} e^{-st} \Big|_4^\infty = -\frac{1}{s} e^{-\infty} + \frac{1}{s} e^{4s} = \frac{1}{s} e^{4s}$$

$$\mathcal{L}[f(t)] = \frac{1}{s} (e^{-2s} - 1) + \frac{1}{s} e^{4s} = \frac{1}{s} (e^{-2s} + e^{4s} - 1)$$

7.

$$f(t) = \begin{cases} 3t & 0 < t < 1 \\ 0 & t \geq 1 \end{cases}$$

$$\mathcal{L}[f(t)] = \int_0^1 e^{-st} 3t dt + \int_1^\infty e^{-st} (0) dt$$

$$\int_0^1 e^{-st} 3t dt = 3 \int_0^1 t e^{-st} dt$$

$$u = t, \quad du = 1$$

$$dv = e^{-st}, \quad v = -\frac{e^{-st}}{s}$$

$$= 3 \left\{ t \left(-\frac{e^{-st}}{s}\right) \Big|_0^1 - \int_0^1 \left(-\frac{e^{-st}}{s}\right) dt = t \left(-\frac{e^{-st}}{s}\right) \Big|_0^1 - \frac{1}{s} \left[\frac{e^{-st}}{s}\right] \right\}$$

$$= \frac{3}{s} \left\{ t(-e^{-st}) \Big|_0^1 - \frac{1}{s} [e^{-st}]_0^1 \right\} = \frac{1}{s} \{ [(-e^{-s}) - 0] - \frac{1}{s} [(e^{-s}) - 1] \}$$

$$\frac{3}{s} [-e^{-s} - \frac{1}{s} e^{-s}] = \frac{3}{s} e^{-s} \left(-1 - \frac{1}{s}\right) + \frac{1}{s^2}$$

$$\int_1^\infty 0 dt = 0$$

$$\mathcal{L}[f(t)] = \frac{3}{s} e^{-s} \left(-1 - \frac{1}{s} + \frac{1}{s}\right)$$

Encontrar $f(t)$ dada su transformada de Laplace $F(s)$, donde $f(t) = \mathcal{L}^{-1}\{F(s)\}$

8.

$$F(s) = \frac{1}{s^2}$$

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s^2}\right] \\ \mathcal{L}^{-1}\left[\frac{n!}{s^{n+1}}\right] &= f(t) = t^n \\ \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] &= t\end{aligned}$$

Usar el teorema de la transformada de la derivada de una funcion para encontrar $F(s)$ dada $f(t)$

9.

$$f(t) = t \sin 3t$$

$$\begin{aligned}\text{sean } f(t) &= t \sin 3t, f(0) = 0 \\ f'(t) &= \sin 3t + 3t \cos 3t, f'(0) = 0 \\ f''(t) &= 3 \cos 3t + 3 \cos 3t - 9t \sin 3t, f''(0) = 6 \\ \mathcal{L}[f''(t)] &= s^2 F(s) - f(0) - f'(0) \\ \mathcal{L}[3 \cos 3t + 3 \cos 3t - 9t \sin 3t] &= s^2 \mathcal{L}[t \sin 3t] - 0 - 0 \\ 6\mathcal{L}[\cos 3t] - 9\mathcal{L}[t \sin 3t] &= s^2 \mathcal{L}[t \sin 3t] \\ 6\mathcal{L}[\cos 3t] &= s^2 \mathcal{L}[t \sin 3t] + 9\mathcal{L}[t \sin 3t] \\ 6\mathcal{L}[\cos 3t] &= (s^2 + 9)\mathcal{L}[t \sin 3t] \\ 6\left[\frac{s}{s^2 + 9}\right] &= (s^2 + 9)\mathcal{L}[t \sin 3t] \\ \mathcal{L}[t \sin 3t] &= \frac{6s}{(s^2 + 9)^2}\end{aligned}$$

10.

$$f(t) = t \cosh t$$

$$\begin{aligned}\text{sean:} \\ f(t) &= t \cosh t, f(0) = 0 \\ f'(t) &= \cosh t + t \sinh t, f'(0) = 1 \\ f''(t) &= \sinh t + \sinh t + t \cosh t \\ \mathcal{L}[f''(t)] &= s^2 F(s) - f(0) - f'(0) \\ \mathcal{L}[2 \sinh t + t \cosh t] &= s^2 \mathcal{L}[t \cosh t] - 0 - 1 \\ 2\mathcal{L}[\sinh t] + \mathcal{L}[t \cosh t] &= s^2 \mathcal{L}[t \cosh t] - 1 \\ 2\mathcal{L}[\sinh t] &= s^2 \mathcal{L}[t \cosh t] - \mathcal{L}[t \cosh t] - 1 \\ 2\frac{1}{s^2 - 1} &= (s^2 - 1)\mathcal{L}[t \cosh t] - 1 \\ \frac{2}{(s^2 - 1)} + 1 &= (s^2 - 1)\mathcal{L}[t \cosh t] \\ \frac{2 + s^2 - 1}{(s^2 - 1)} &= (s^2 - 1)\mathcal{L}[t \cosh t] \\ \mathcal{L}[t \cosh t] &= \frac{s^2 + 1}{(s^2 - 1)^2}\end{aligned}$$

11.

$$f(t) = t^2 \cos 3t$$

Sean:

$$f(t) = t^2 \cos 3t, f(0) = 0$$

$$f'(t) = 2t \cos 3t - 3t^2 \sin 3t, f'(0) = 0$$

$$f''(t) = 2 \cos 3t - 6t \sin 3t - (6t \sin 3t + 9t^2 \cos 3t) = 2 \cos 3t - 12t \sin 3t - 9t^2 \cos 3t$$

Aplicando el teorema

$$\mathcal{L}[f''(t)] = s^2 F(s) - f(0) - f'(0)$$

$$\mathcal{L}[2 \cos 3t - 12t \sin 3t - 9t^2 \cos 3t] = s^2 \mathcal{L}[t^2 \cos 3t] - 0 - 0$$

$$2\mathcal{L}[\cos 3t] - 12\mathcal{L}[t \sin 3t] - 9\mathcal{L}[t^2 \cos 3t] = s^2 \mathcal{L}[t^2 \cos 3t]$$

$$2\mathcal{L}[\cos 3t] - 12\mathcal{L}[t \sin 3t] = s^2 \mathcal{L}[t^2 \cos 3t] + 9\mathcal{L}[t^2 \cos 3t]$$

$$2 \frac{s}{s^2 + 9} - 12 \frac{6s}{(s^2 + 9)^2} = (s^2 + 9) \mathcal{L}[t^2 \cos 3t] \Leftarrow \text{podemos observar que la transformada de } t \sin 3t \text{ es } \frac{6s}{(s^2 + 9)^2} \text{ pues ya la habiamos resuelto anteriormente.}$$

$$\frac{2s(s^2 + 9) - 36s}{(s^2 + 9)^2} = (s^2 + 9) \mathcal{L}[t^2 \cos 3t]$$

$$\frac{2s^3 + 18s - 36s}{(s^2 + 9)^2} = (s^2 + 9) \mathcal{L}[t^2 \cos 3t]$$

$$\frac{2s^3 - 18s}{(s^2 + 9)^2(s^2 + 9)} = \mathcal{L}[t^2 \cos 3t]$$

$$\mathcal{L}[t^2 \cos 3t] = \frac{2s^3 - 18s}{(s^2 + 9)^3}$$

. Hallar $f(t)$ mediante el teorema de la transformada de la integral dada $F(s)$

12.

$$F(s) = \frac{1}{s(s-4)}$$

$$\mathcal{L}^{-1} \left[\frac{1}{s(s-4)} \right]$$

$$\text{Sabemos que } \mathcal{L}^{-1} \left[\frac{1}{s-a} \right] = e^{at}, \text{ entonces:}$$

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s(s-4)}\right] &= \int_0^t e^{a\tau} d\tau = \frac{e^{a\tau}}{a} \Big|_0^t = \frac{e^{at}}{a} - \frac{1}{a} \text{ donde: } a = 4 \\ &= \frac{e^{4t}}{4} - \frac{1}{4} = \frac{1}{4}(e^{4t} - 1)\end{aligned}$$

13.

$$F(s) = \frac{1}{s^2(s+3)}$$

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s^2(s+3)}\right] &= \\ \text{Si } \mathcal{L}^{-1}\left[\frac{1}{(s+a)}\right] &= e^{at}, \text{ donde } a = 3, \text{ entonces:} \\ \mathcal{L}^{-1}\left[\frac{1}{s(s+3)}\right] &= \int_0^t e^{3\tau} d\tau = \frac{1}{3}e^{3\tau} \Big|_0^t = \frac{1}{3}e^{3t} - \frac{1}{3} \\ \mathcal{L}^{-1}\left[\frac{1}{s^2(s+3)}\right] &= \int_0^t \left(\frac{1}{3}e^{3t} - \frac{1}{3}\right) d\tau = \left[\frac{1}{9}e^{3\tau} - \frac{1}{3}\tau\right]_0^t = \frac{1}{9}e^{3t} - \frac{1}{3}t - \frac{1}{9} \\ f(t) &= \frac{1}{9}(e^{3t} - 3t - 1)\end{aligned}$$

14.

$$F(s) = \frac{3}{s^2(s^2-9)}$$

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{3}{s^2(s^2-9)}\right] &= \\ \text{Conociendo } \mathcal{L}^{-1}\left[\frac{a}{s^2-a^2}\right] &= \sinh at, \text{ donde } a = 3, \text{ entonces:} \\ \mathcal{L}^{-1}\left[\frac{3}{s(s^2-9)}\right] &= \int_0^t \sinh 3\tau d\tau = \frac{1}{3} \cosh 3\tau \Big|_0^t = \frac{1}{3} \cosh 3t - \frac{1}{3} \\ \mathcal{L}^{-1}\left[\frac{3}{s^2(s^2-9)}\right] &= \int_0^t \left(\frac{1}{3} \cosh 3\tau - \frac{1}{3}\right) d\tau = \left[\frac{1}{9} \sinh 3\tau - \frac{1}{3}\tau\right]_0^t = \frac{1}{9} \sinh 3t - \\ &\frac{1}{3}t \\ f(t) &= \frac{1}{9}(\sinh 3t - 3t)\end{aligned}$$

Resolver las siguientes ecuaciones diferenciales, usando la transformada de Laplace

15.

$$y'' + y = 0 \quad , \quad y(0) = 1$$

$$\mathcal{L}\{y'' + y\} = \mathcal{L}\{0\}$$

Por el teorema de transformadas de derivadas

$$s^2 Y(s) - sy(0) - y'(0) + Y(s) = 0$$

Sustituyendo el valor inicial

$$s^2 Y(s) - s(1) - 0 + Y(s) = 0$$

$$Y(s)(s^2 + 1) - 1 = 0$$

$$Y(s) = \frac{1}{s^2 + 1}$$

Aplicamos la transformada inversa a $Y(s)$

$$\mathcal{L}^{-1} \left[\frac{1}{s^2 + 1} \right] = e^{-t}$$

por linealidad

$$y = e^{-x}$$

16.

$$y'' + 4y = 2, \quad y(0) = 0, \quad y'(0) = 0$$

$$\mathcal{L}[y'' + 4y] = \mathcal{L}[2]$$

$$s^2 Y(s) - sy(0) - y'(0) + 4Y(s) = \frac{2}{s}$$

Sustituyendo el valor inicial

$$s^2 Y(s) - 0 - 0 + 4Y(s) = \frac{2}{s}$$

$$Y(s)(s^2 + 4) = \frac{2}{s}$$

$$Y(s) = \frac{2}{s(s^2 + 4)}$$

se aplica la transformada inversa

$$\mathcal{L}^{-1} \left[\frac{2}{s(s^2 + 4)} \right] =$$

$$\text{Si } \mathcal{L}^{-1} \left[\frac{\omega}{s^2 + \omega^2} \right] = \sin \omega t \text{ donde } \omega = 2, \text{ entonces}$$

$$\mathcal{L}^{-1} \left[\frac{2}{s(s^2 + 4)} \right] = \int_0^t \sin 2\tau d\tau = -\frac{1}{2} \cos 2\tau \Big|_0^t = -\frac{1}{2} \cos 2t + \frac{1}{2}$$

$$f(t) = \frac{1}{2} - \frac{1}{2} \cos 2t$$

por linealidad

$$y = \frac{1}{2} - \frac{1}{2} \cos 2x$$

17.

$$y'' + 16y = 4, \quad y(0) = 1, \quad y'(0) = 0$$

Aplicamos transformada de Laplace a la ecuación

$$\mathcal{L}[y'' + 16y] = \mathcal{L}[4]$$

$$s^2Y(s) + sy(0) + y'(0) + 16Y(s) = \frac{4}{s}$$

Sustituimos los valores iniciales, y despejamos $Y(s)$

$$Y(s)(s^2 + 16) + s + 0 = \frac{4}{s}$$

$$Y(s) = \frac{\frac{4}{s} - s}{(s^2 + 16)} = \frac{\frac{4-s^2}{s}}{(s^2 + 16)} = \frac{4-s^2}{s(s^2 + 16)}$$

Aplicamos transformada inversa

$$\mathcal{L}^{-1} \left[\frac{4-s^2}{s(s^2 + 16)} \right] = \mathcal{L}^{-1} \left[\frac{4}{s(s^2 + 16)} \right] - \mathcal{L}^{-1} \left[\frac{s}{(s^2 + 16)} \right] =$$

$$\mathcal{L}^{-1} \left[\frac{s}{s^2 + \omega^2} \right] = \cos \omega t, \text{ donde } \omega = 4$$

$$\mathcal{L}^{-1} \left[\frac{s}{(s^2 + 16)} \right] = \cos 4t$$

$$\text{Entonces para } \mathcal{L}^{-1} \left[\frac{4}{s(s^2 + 16)} \right]$$

$$\text{si } \mathcal{L}^{-1} \left[\frac{\omega}{s^2 + \omega^2} \right] = \sin \omega t, \text{ donde } \omega = 4$$

$$\mathcal{L}^{-1} \left[\frac{4}{s(s^2 + 16)} \right] = \int_0^t \sin 4\tau d\tau = - \left[\frac{1}{4} \cos 4\tau \right]_0^t = - \left[\frac{1}{4} \cos 4t - \frac{1}{4} \right] =$$

$$-\frac{1}{4} \cos 4t + \frac{1}{4}$$

por linealidad

$$y = \frac{3}{4} \cos 4x + \frac{1}{4}$$

18.

$$y''' - 2y' + 5y = 0, \quad y(0) = 2, \quad y'(0) = 4;$$

Aplicamos la transformada de Laplace a la ecuación

$$\mathcal{L}[y''' - 2y' + 5y] = \mathcal{L}[0]$$

Usamos formulas de transformadas de Laplace de derivadas

$$\mathcal{L}[y](s) = Y(s)$$

$$\mathcal{L}[y'](s) = sY(s) - y(0) = sY(s) - 2$$

$$\mathcal{L}[y''](s) = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - 2s - 4$$

Al sustituir estas expresiones en la ecuación nos da

$$s^2Y(s) - 2s - 4 - 2[sY(s) - 2] + 5Y(s) = 0$$

$$s^2Y(s) - 2s - 4 - 2sY(s) - 4 + 5Y(s) = 0$$

Factorizamos y despejamos $Y(s)$

$$Y(s)(s^2 - 2s + 5) - 2s - 8 = 0$$

$$Y(s)(s^2 - 2s + 5) = 2s + 8$$

$$Y(s) = \frac{2s + 8}{(s^2 - 2s + 5)}$$

Ahora calculamos la transformada inversa de la función racional $Y(s)$, para esto usaremos un desarrollo en fracciones parciales

Como $s^2 - 2s + 5$ es irreducible, escribimos este factor en la forma $(s - \alpha)^2 + \beta^2$
 $(s - 1)^2 + 2^2$

$$Y(s) = \frac{2s + 8}{(s - 1)^2 + 2^2} = \frac{A(s - 1) + 2B}{(s - 1)^2 + 4}$$

$$2s + 8 = A(s - 1) + 2B$$

Hacemos $s = -1, 1$

en el primer caso obtenemos

$$2(-1) + 8 = A(-2) + 2B, 6 = -2A + 2B$$

$$2(1) + 8 = A(0) + 2B, B = 5$$

$$6 = -2A + 10, A = 2$$

Sustituimos estos valores en el desarrollo de fracciones parciales.

$$Y(s) = \frac{2(s - 1) + 2(5)}{(s - 1)^2 + 4} = \frac{2s - 2 + 10}{(s - 1)^2 + 4} = \frac{2s + 8}{(s - 1)^2 + 4}$$

aplicamos transformada inversa de Laplace

$$2\mathcal{L}^{-1}\left[\frac{s}{(s - 1)^2 + 4}\right] + 4\mathcal{L}^{-1}\left[\frac{2}{(s - 1)^2 + 4}\right]$$

$$f(t) = 2e^t \cos 2t + 4e^t \sin 2t$$

19.

$$w'' + w = t^2 + 2, \quad w(0) = 1, \quad w'(0) = -1$$

$$s^2 W(s) - sw(0) - w'(0) + W(s) = \mathcal{L}[t^2] + \mathcal{L}[2]$$

$$W(s)(s^2 + 1) - s + 1 = \frac{2}{s^3} + \frac{2}{s}$$

$$W(s)(s^2 + 1) = \frac{2 + 2s^2}{s^3} + s - 1$$

$$W(s) = \frac{\frac{2(1 + s^2) + s^4 - s^3}{s^3}}{s^2 + 1} = \frac{2 + 2s^2 + s^4 - s^3}{s^3(s^2 + 1)}$$

Aplicamos la transformada inversa

$$f(t) = \mathcal{L}^{-1}\left[\frac{2}{s^3(s^2 + 1)}\right] + \mathcal{L}^{-1}\left[\frac{2}{s(s^2 + 1)}\right] + \mathcal{L}^{-1}\left[\frac{s}{(s^2 + 1)}\right] - \mathcal{L}^{-1}\left[\frac{1}{(s^2 + 1)}\right]$$

Aplicamos el teorema de la transformada de la integral para las dos primeras transformadas inversas.

$$\mathcal{L}^{-1}\left[\frac{2}{s^3(s^2 + 1)}\right] = \int_0^t 2 \sin \tau d\tau = 2 \cos \tau|_0^t = -2 \cos t + 2$$

$$= -2 \int_0^t [\cos \tau - 1] d\tau = -2(\sin \tau - \tau)|_0^t = -2(\sin t - t) = 2t - 2 \sin t$$

$$= 2 \int_0^t (\tau - \sin \tau) d\tau = 2\left(\frac{1}{2}\tau^2 + \cos \tau\right)|_0^t = t^2 + 2 \cos t - 1$$

$$= \int_0^t (\tau^2 + 2 \cos \tau - 1) d\tau = \left(\frac{1}{3}\tau^3 + 2 \sin \tau - \tau\right)|_0^t = \frac{1}{3}t^3 + 2 \sin t - t$$

$$\mathcal{L}^{-1}\left[\frac{2}{s(s^2 + 1)}\right] = \int_0^t 2 \sin \tau d\tau = -2 \cos \tau|_0^t = -2 \cos t + 2$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s^2 + 1)}\right] = \cos t$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s^2+1)}\right] = \sin t$$

Sustituyendo

$$f(t) = \frac{1}{3}t^3 + 2\sin t - t - 2\cos t + 2 + \cos t - \sin t$$

References

- [1] Ecuaciones diferenciales con aplicaciones de modelado, Dennis G. Zill, Novena edición, Editorial Cengage learning. Ejercicios 7.1, Problema 21
- [2] Ecuaciones diferenciales, Isabel Carmona Jover, Cuarta edición, Editorial Pearson Educación, Ejercicios 7.1, Problemas 1, 4, 14, 18. Ejercicios 7.2, Problemas 1, 2, 5, 9, 10, 15, 18, 19
- [3] Ecuaciones diferenciales y problemas con valores en la frontera, R. Kent Nagle, Edward B. Saff, Arthur David Snider, Cuarta edición, Editorial Pearson educación. Ejercicios 7.5, Problemas 1